

BFS Traversal

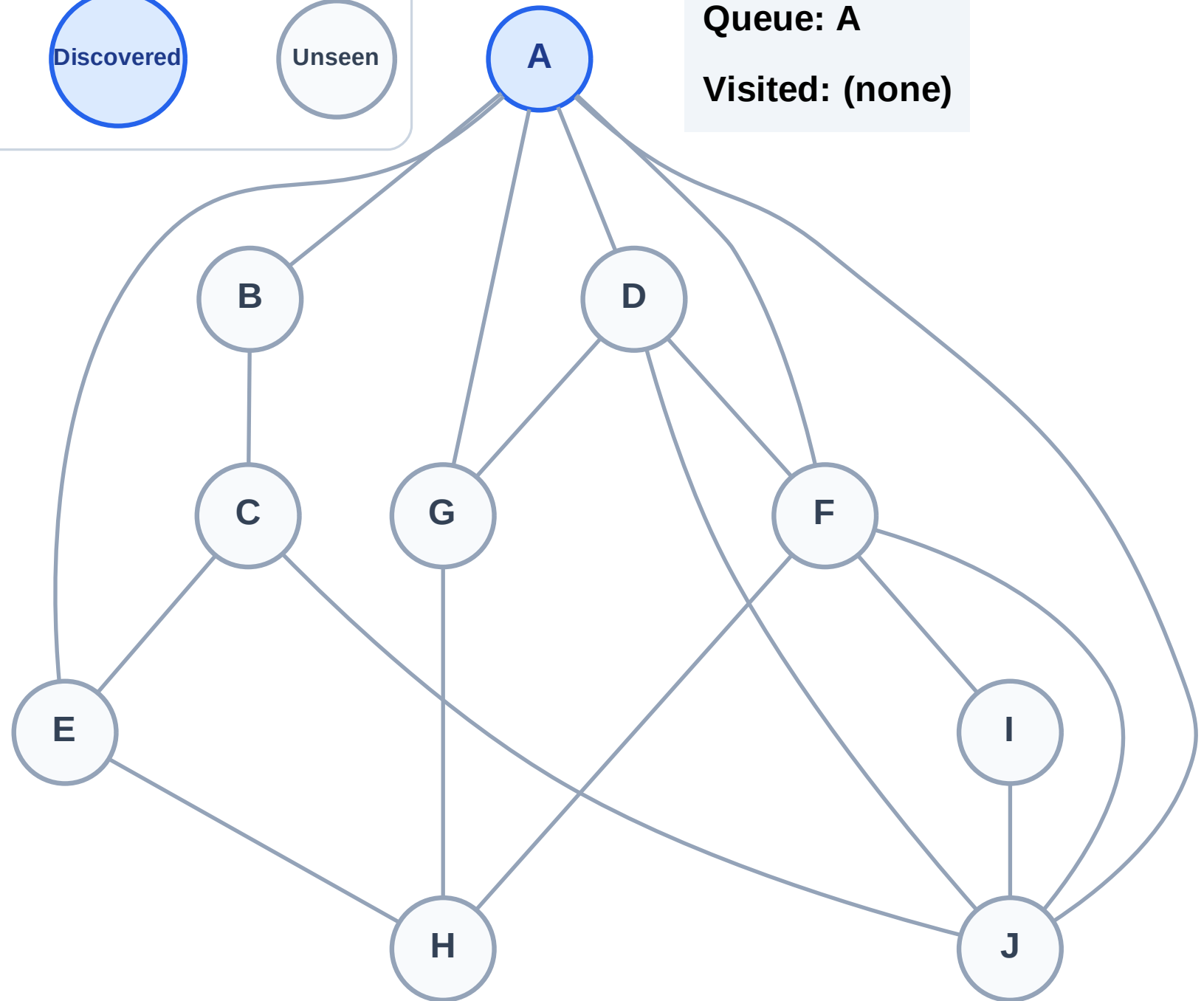
Step 1: Start at A

Legend



Queue: A

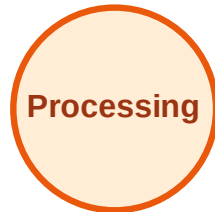
Visited: (none)



BFS Traversal

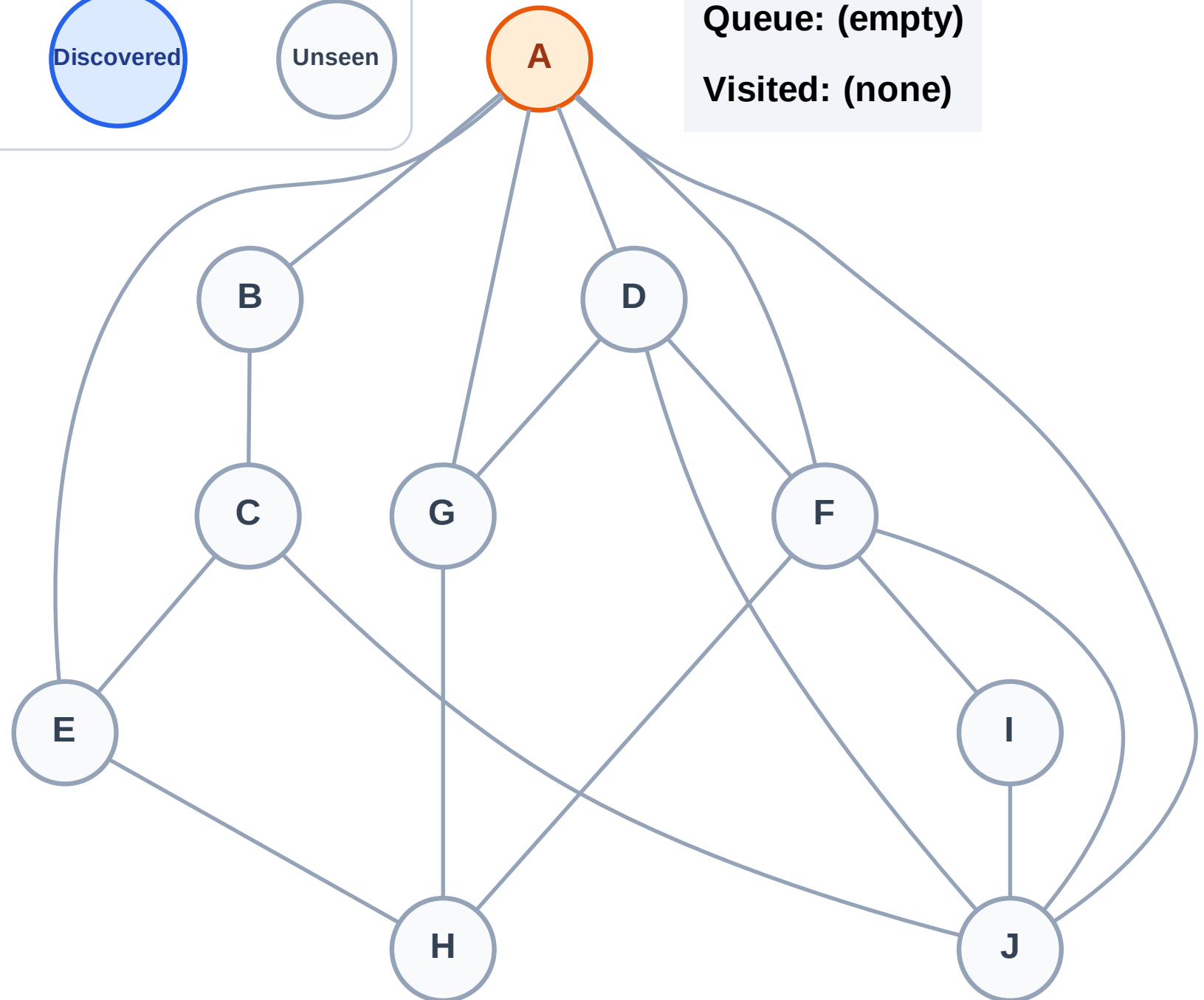
Step 2: Process node A

Legend



Queue: (empty)

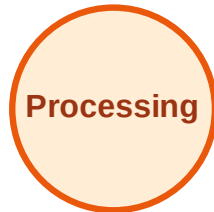
Visited: (none)



BFS Traversal

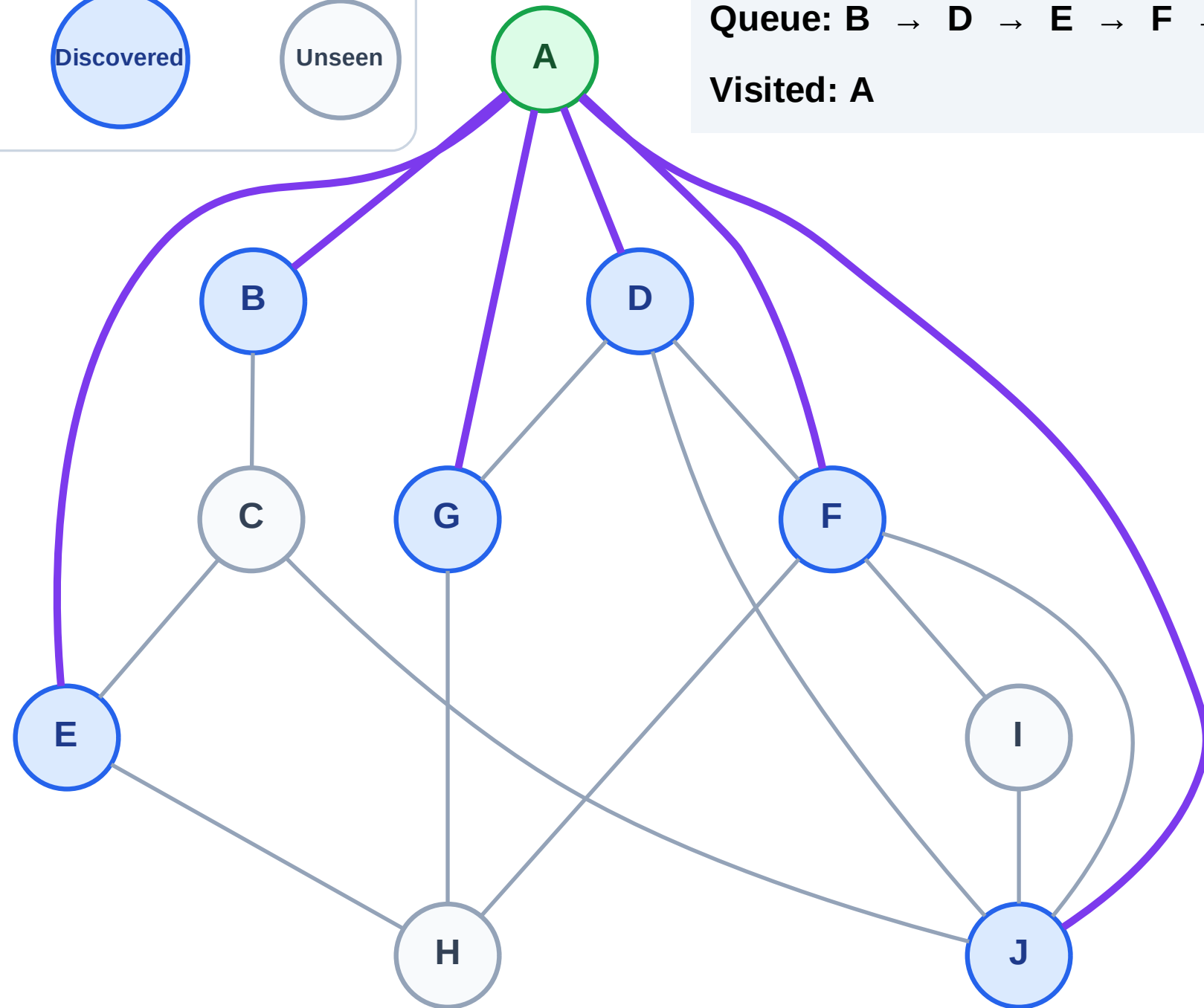
Step 3: Discover B, D, E, F, G, J from A

Legend



Queue: B → D → E → F → G → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

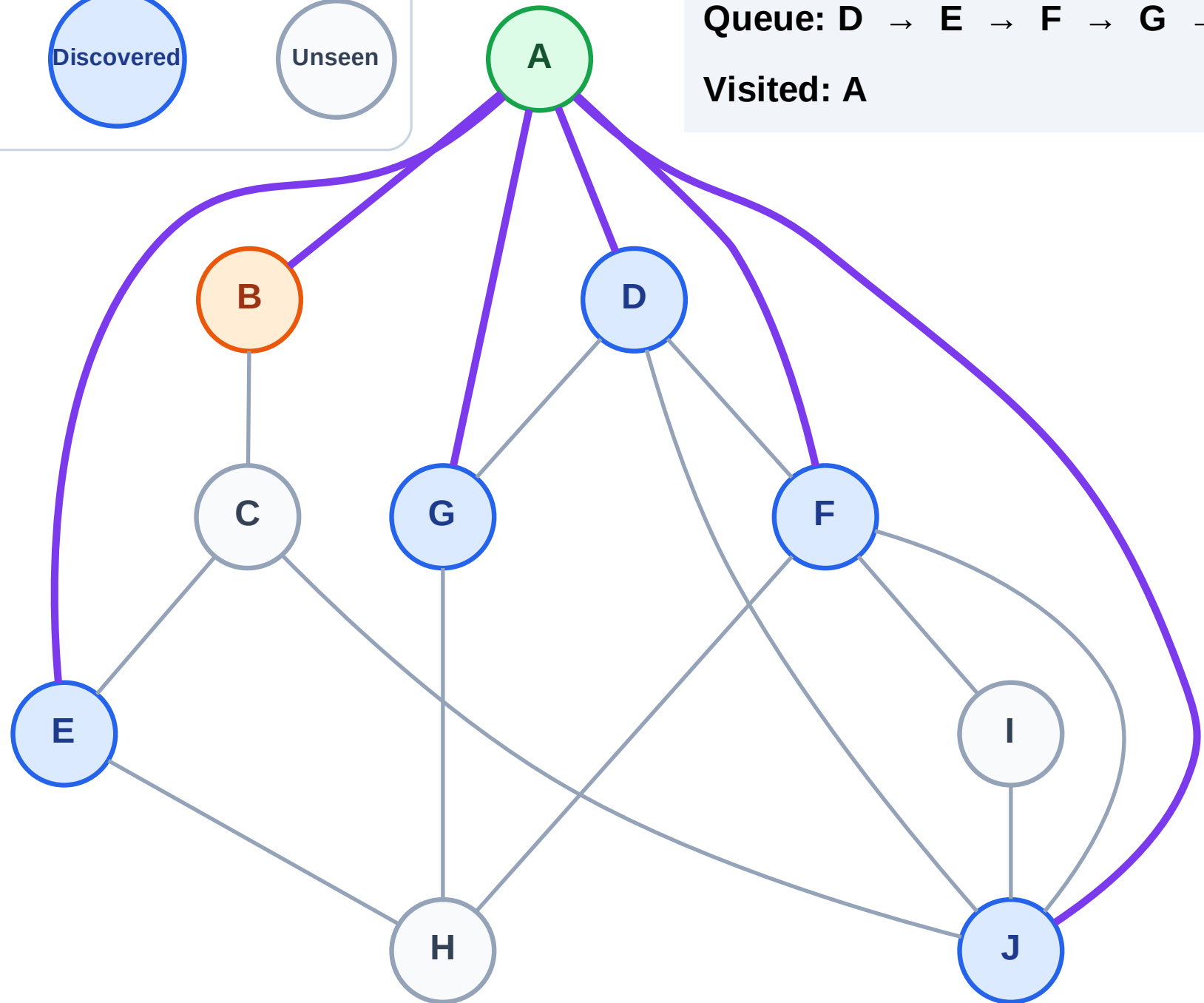
Processing

Discovered

Unseen

Queue: D → E → F → G → J

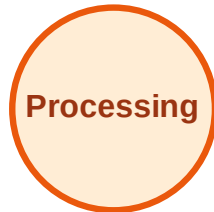
Visited: A



BFS Traversal

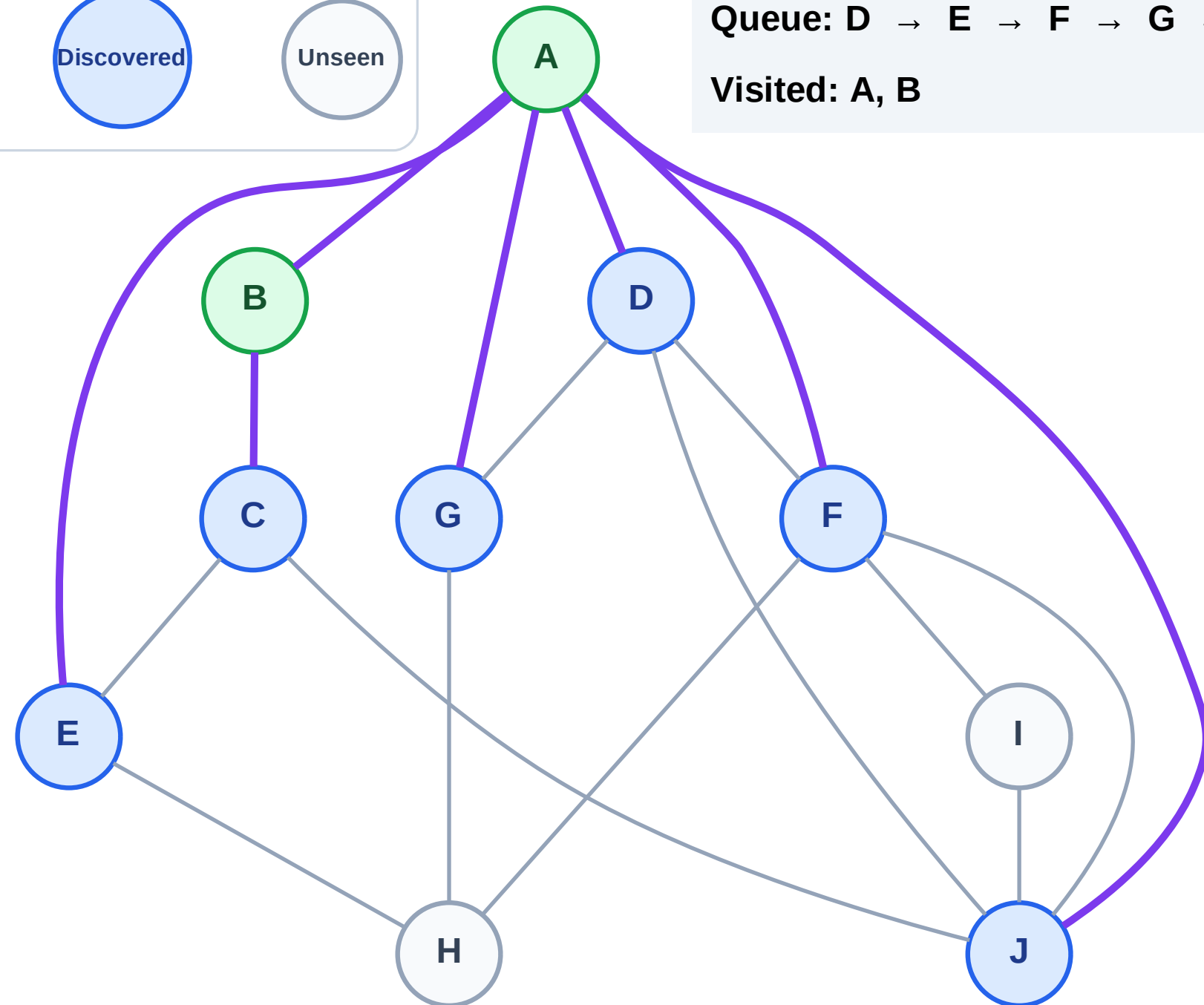
Step 5: Discover C from B

Legend



Queue: D → E → F → G → J → C

Visited: A, B



BFS Traversal

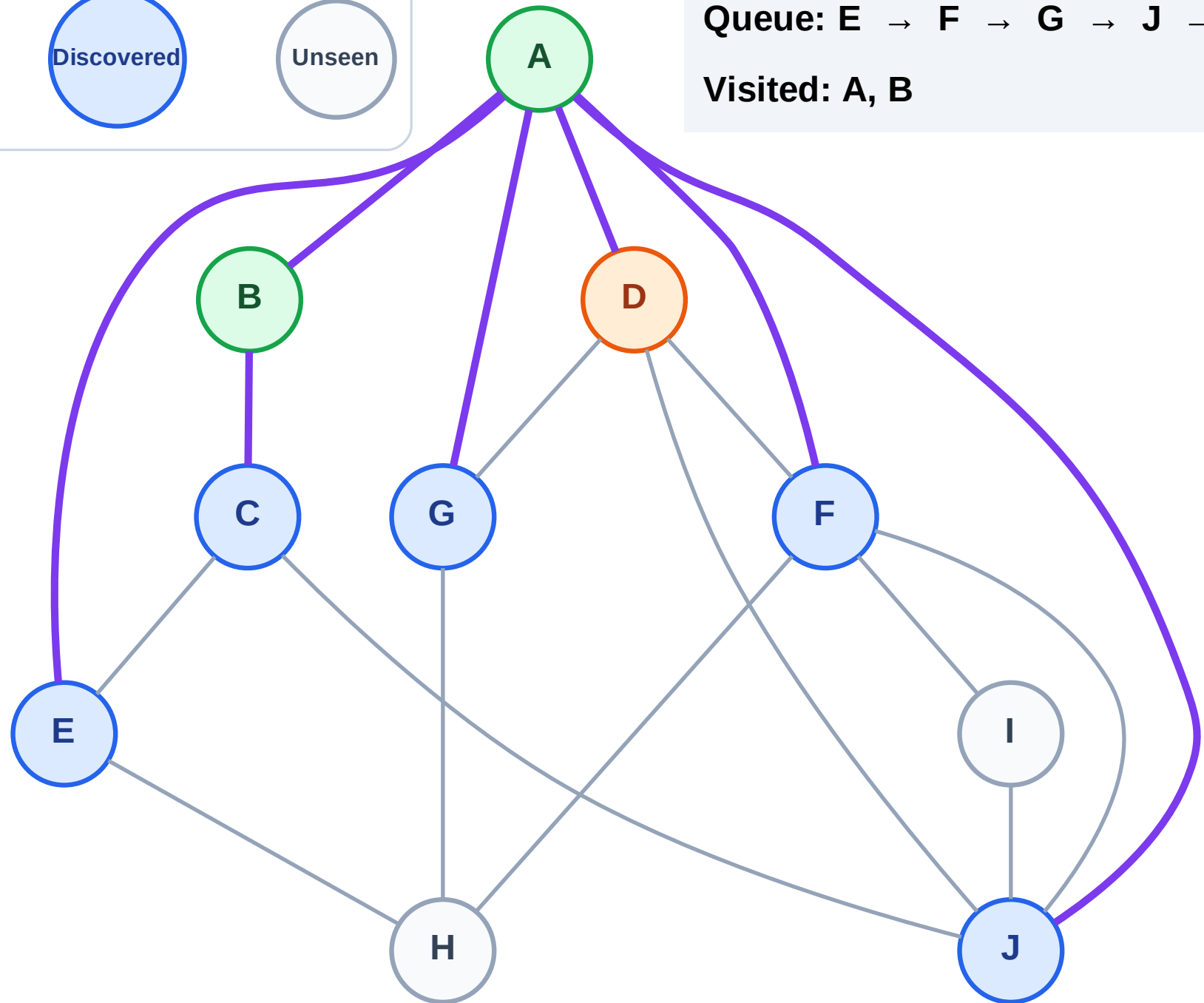
Step 6: Process node D

Legend



Queue: E → F → G → J → C

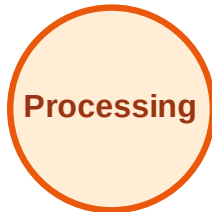
Visited: A, B



BFS Traversal

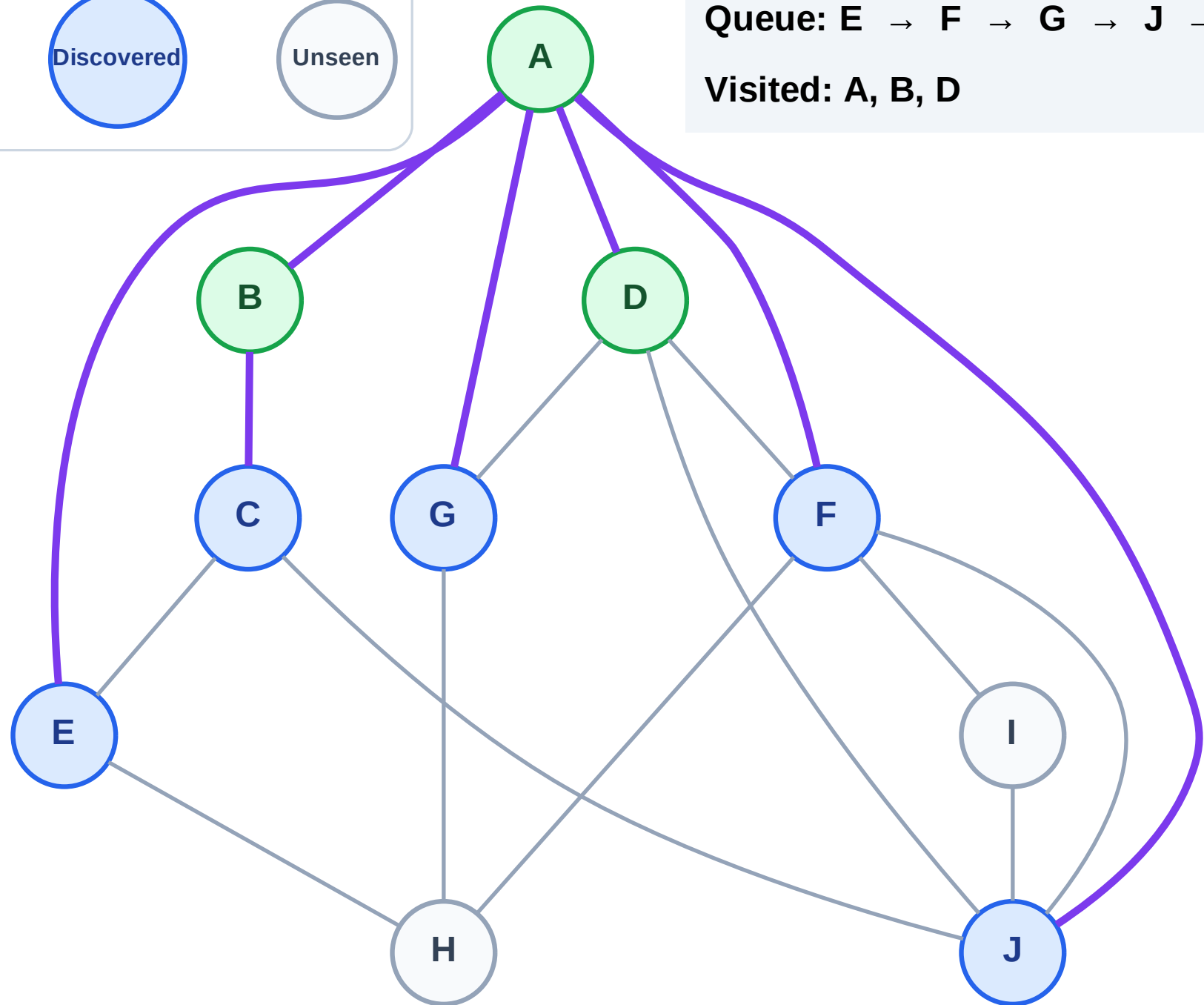
Step 7: No new neighbors from D

Legend



Queue: E → F → G → J → C

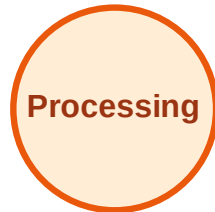
Visited: A, B, D



BFS Traversal

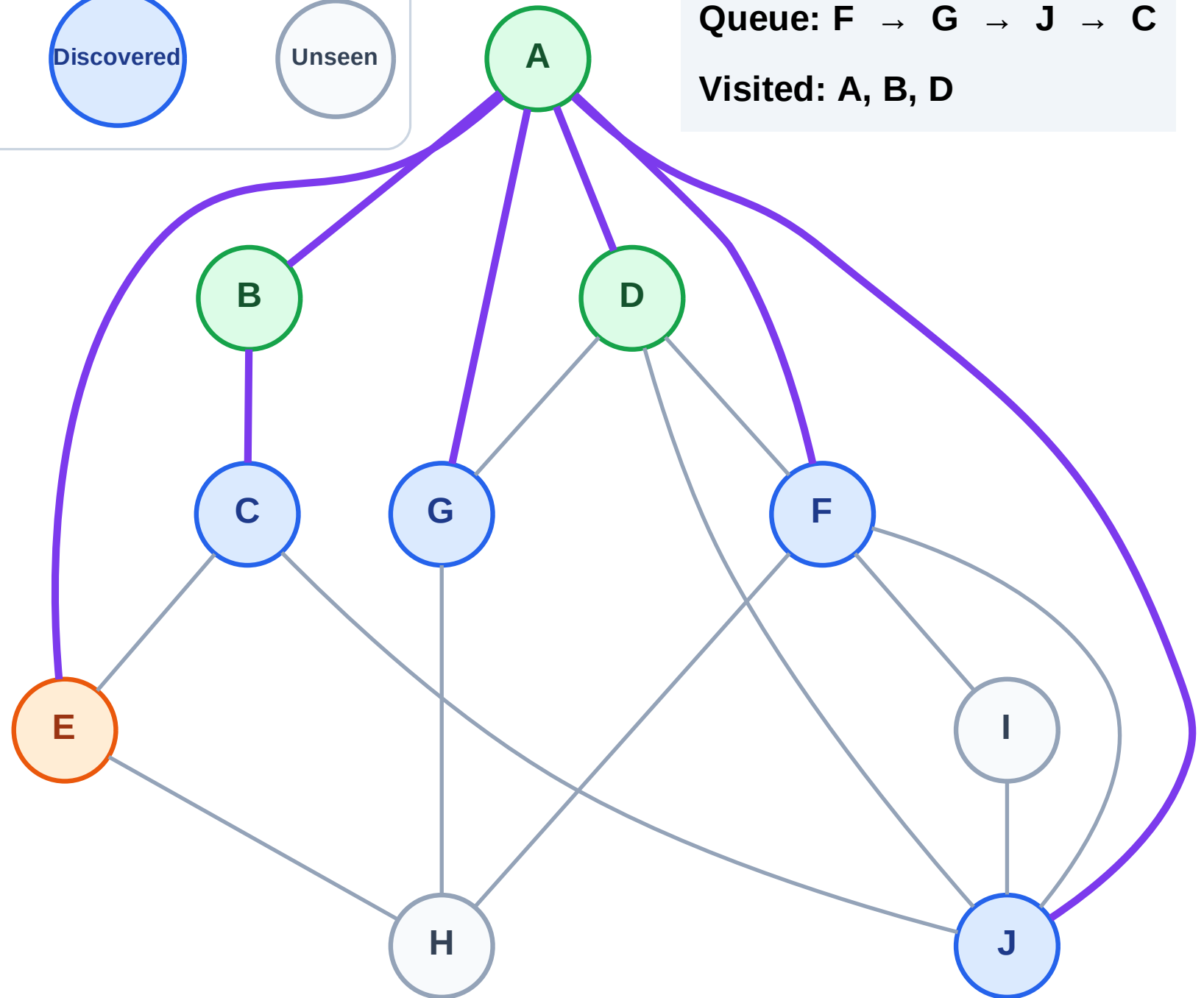
Step 8: Process node E

Legend



Queue: F → G → J → C

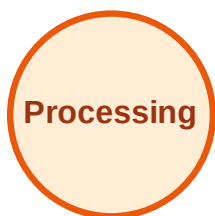
Visited: A, B, D



BFS Traversal

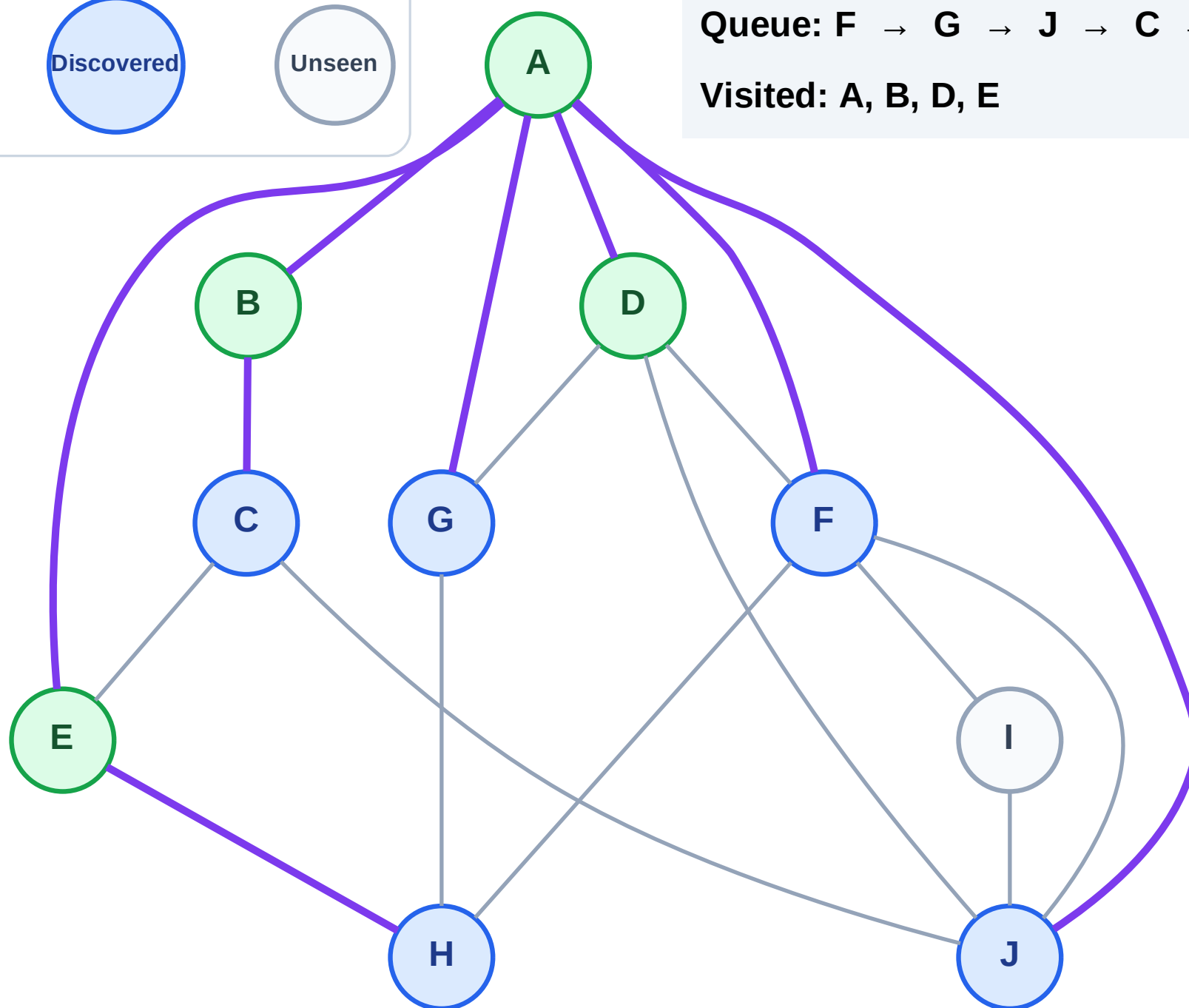
Step 9: Discover H from E

Legend



Queue: F → G → J → C → H

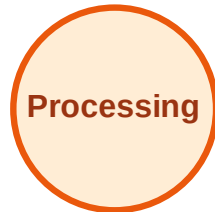
Visited: A, B, D, E



BFS Traversal

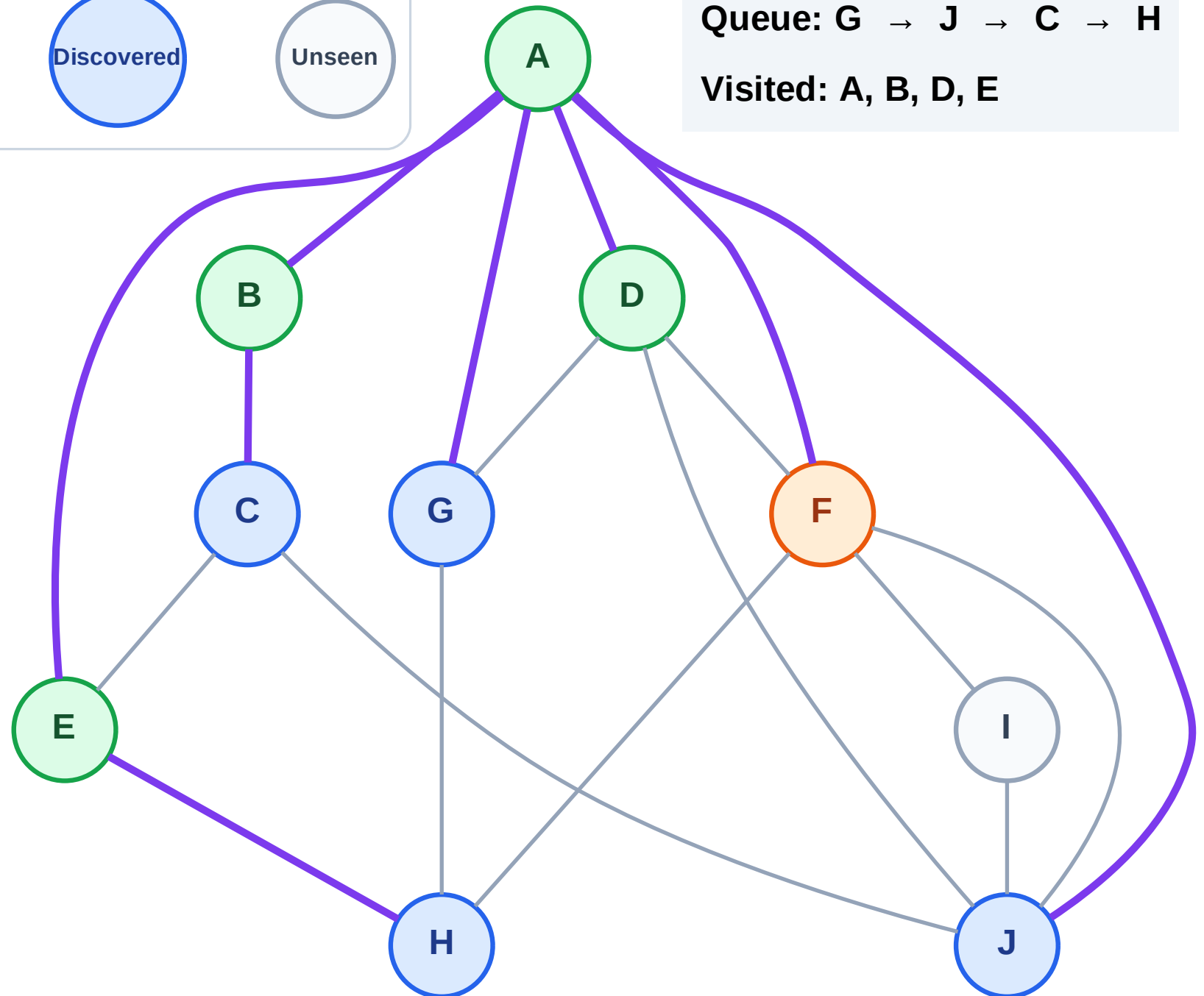
Step 10: Process node F

Legend



Queue: G → J → C → H

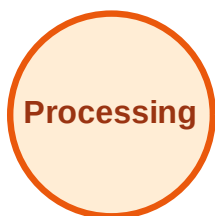
Visited: A, B, D, E



BFS Traversal

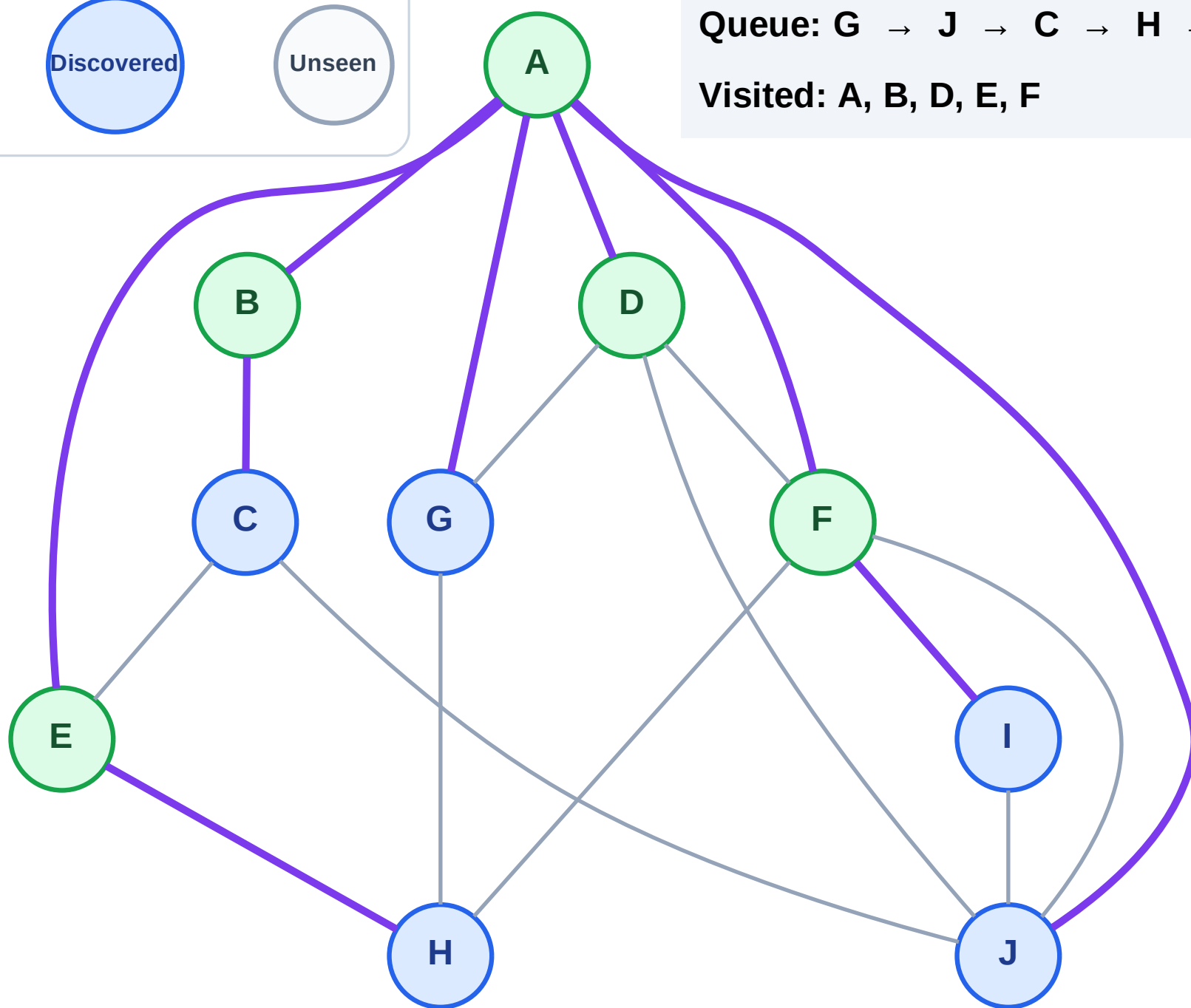
Step 11: Discover I from F

Legend



Queue: G → J → C → H → I

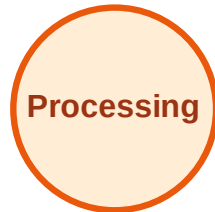
Visited: A, B, D, E, F



BFS Traversal

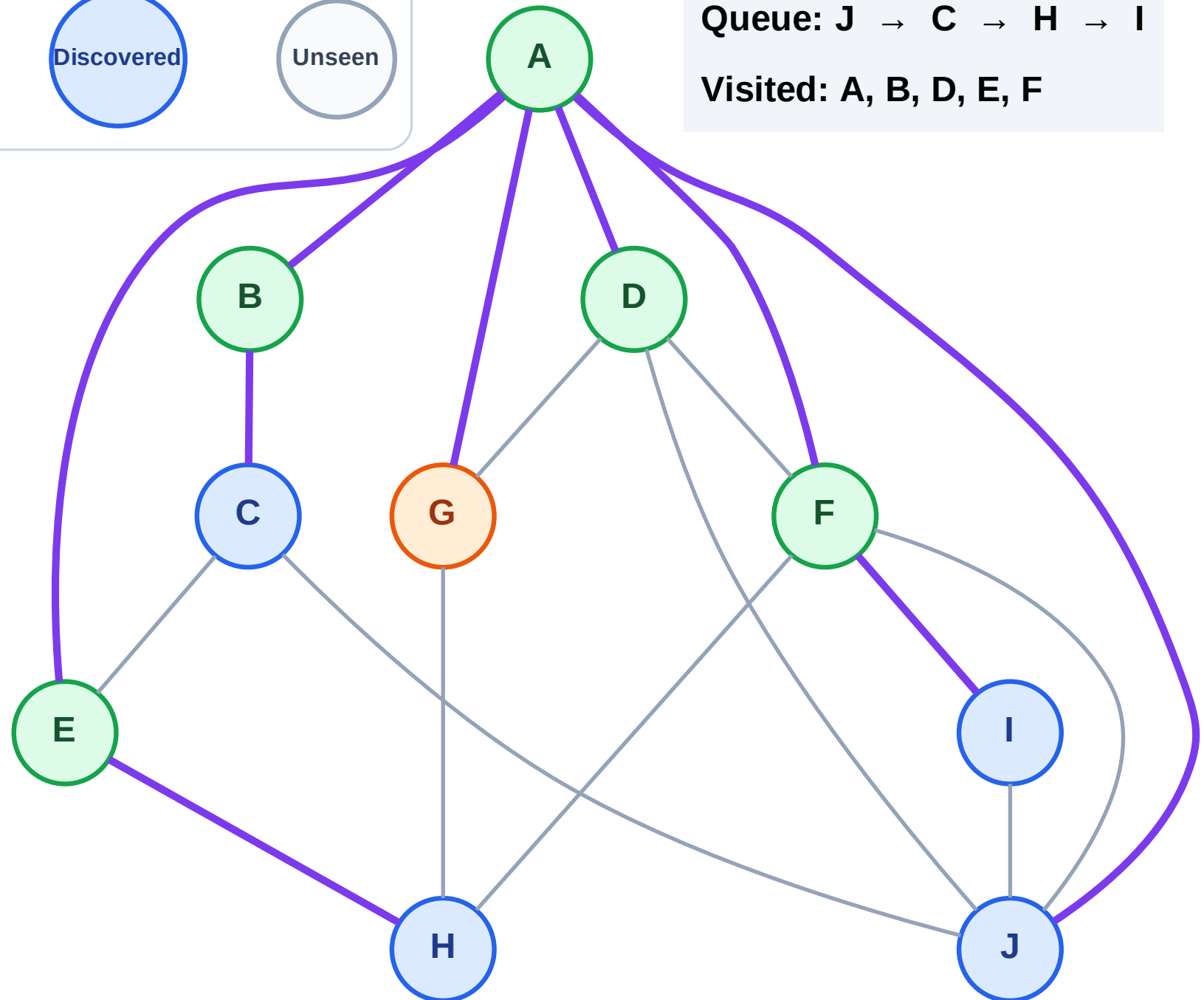
Step 12: Process node G

Legend



Queue: J → C → H → I

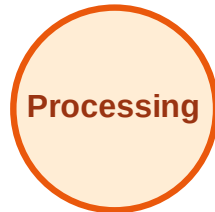
Visited: A, B, D, E, F



BFS Traversal

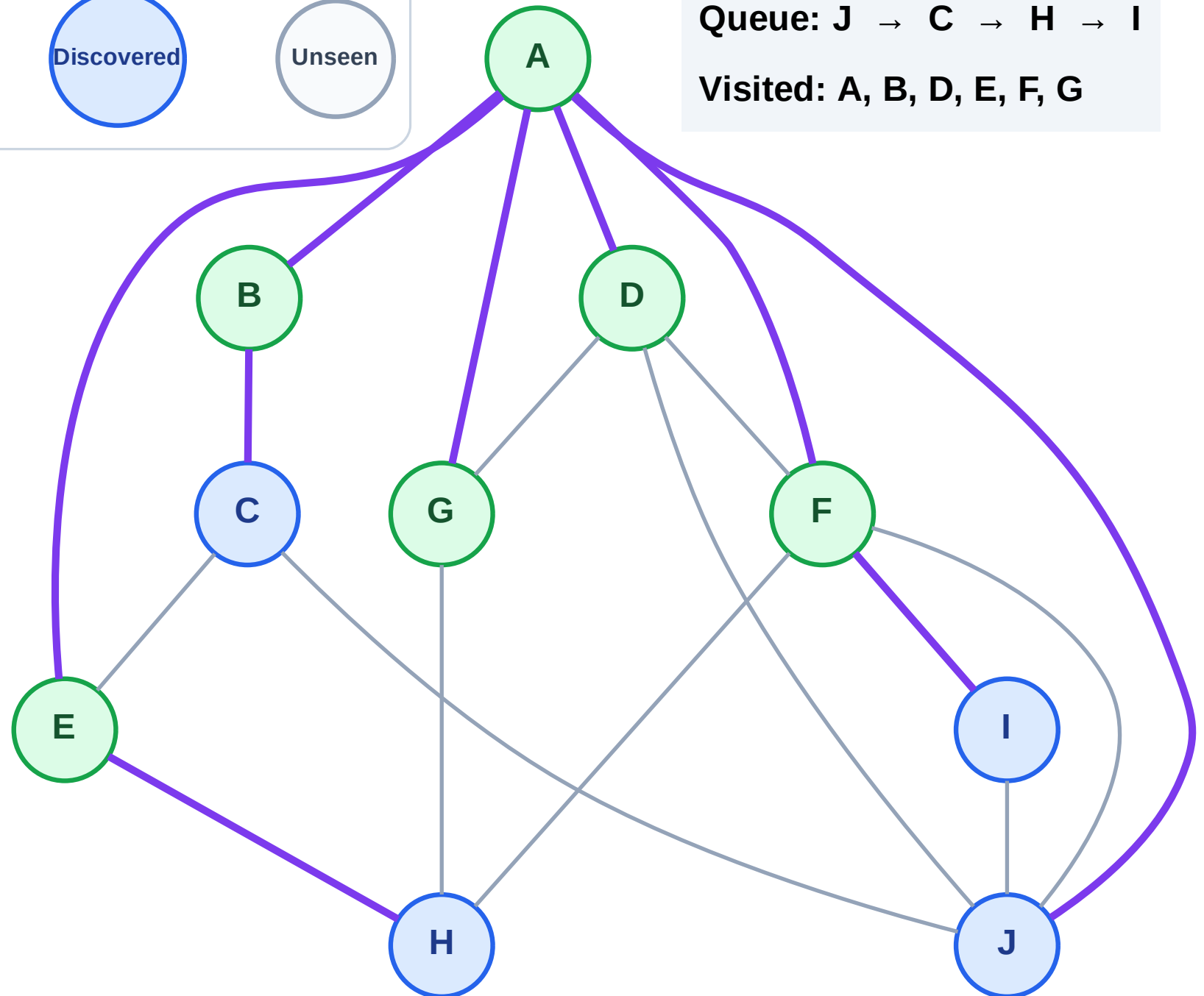
Step 13: No new neighbors from G

Legend



Queue: J → C → H → I

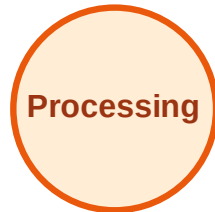
Visited: A, B, D, E, F, G



BFS Traversal

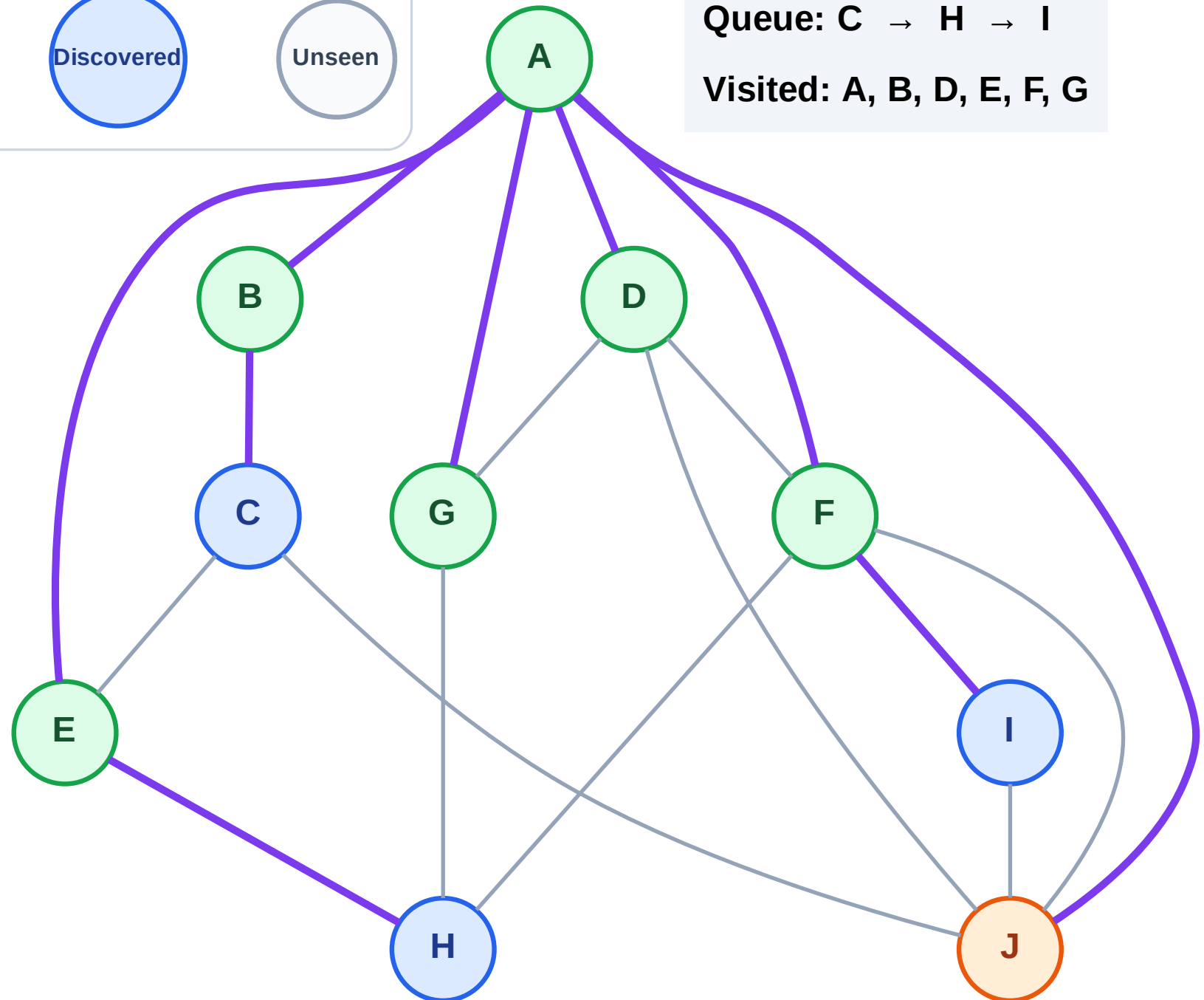
Step 14: Process node J

Legend



Queue: C → H → I

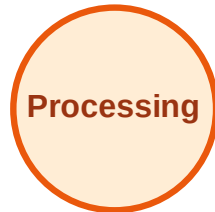
Visited: A, B, D, E, F, G



BFS Traversal

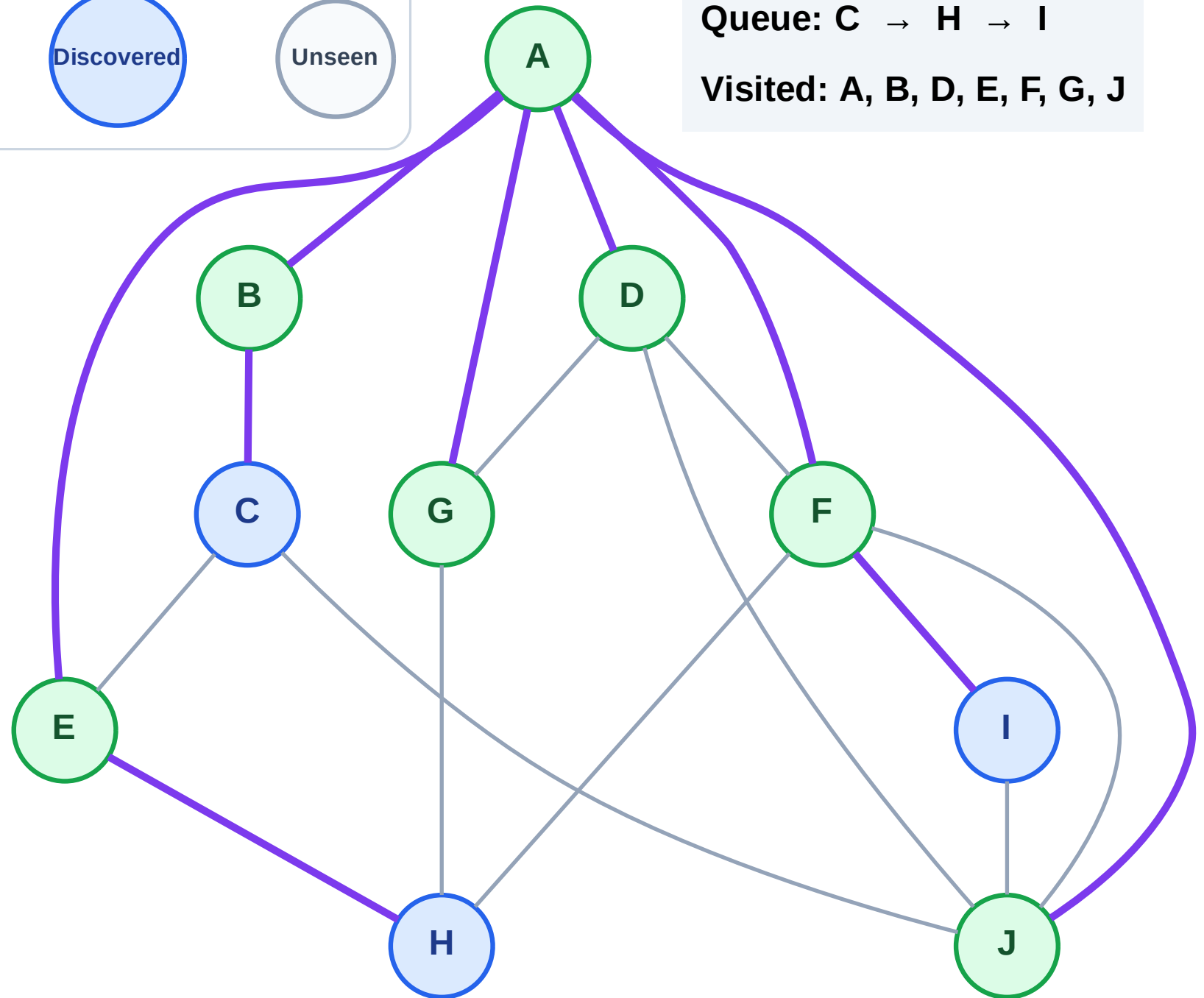
Step 15: No new neighbors from J

Legend



Queue: C → H → I

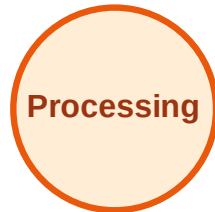
Visited: A, B, D, E, F, G, J



BFS Traversal

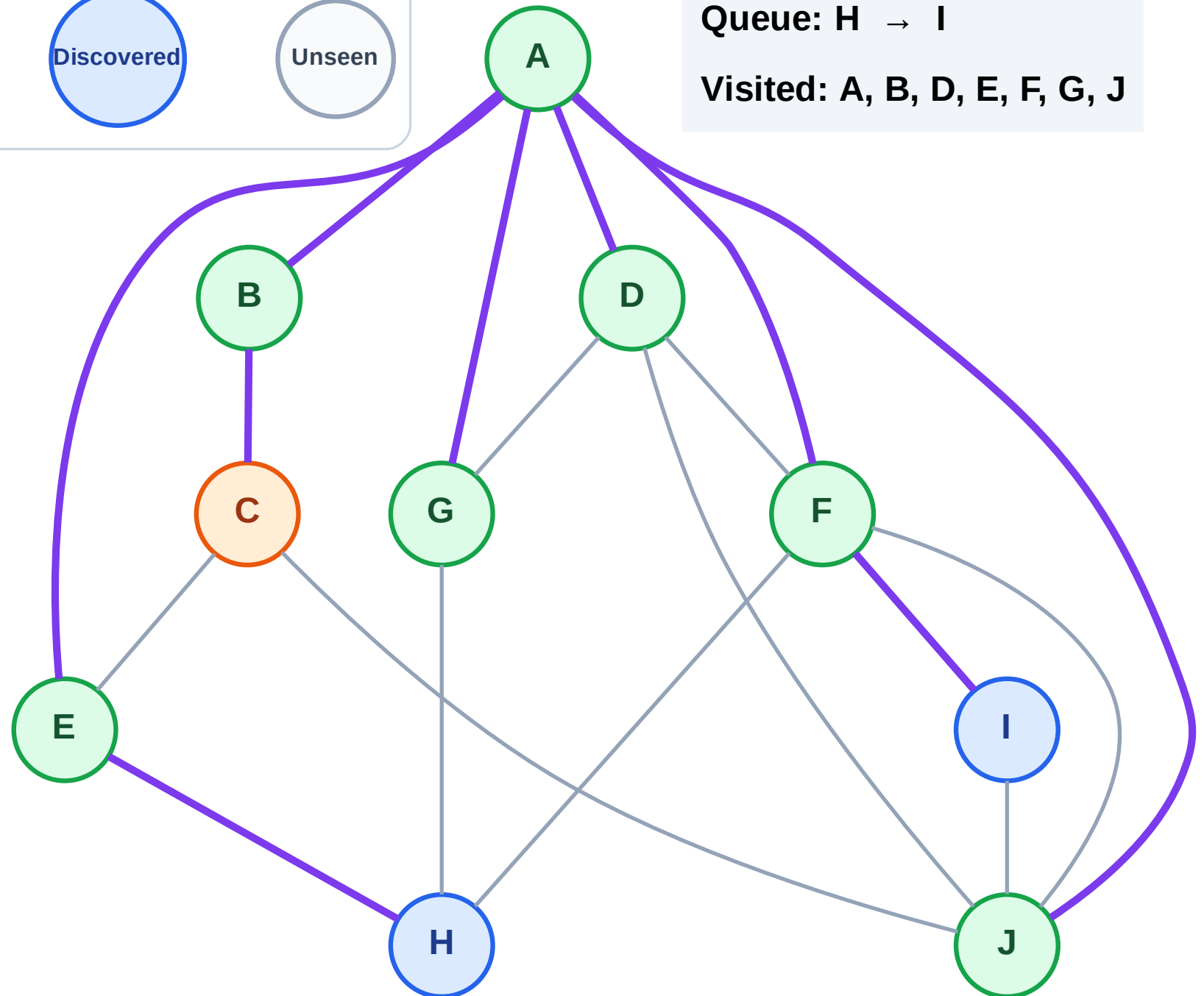
Step 16: Process node C

Legend



Queue: H → I

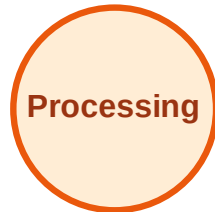
Visited: A, B, D, E, F, G, J



BFS Traversal

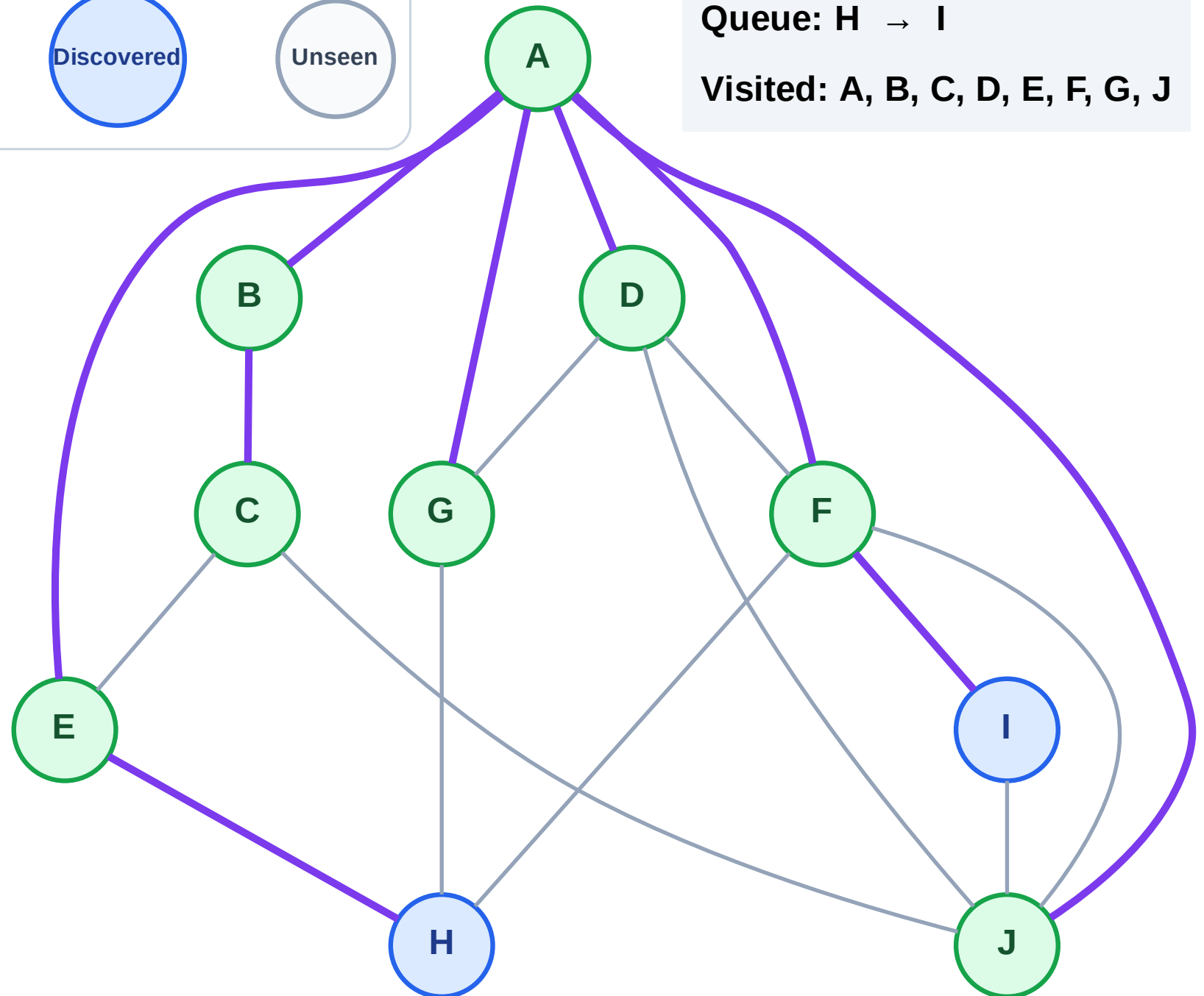
Step 17: No new neighbors from C

Legend



Queue: H → I

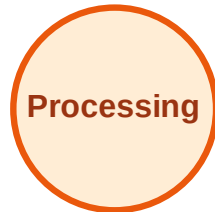
Visited: A, B, C, D, E, F, G, J



BFS Traversal

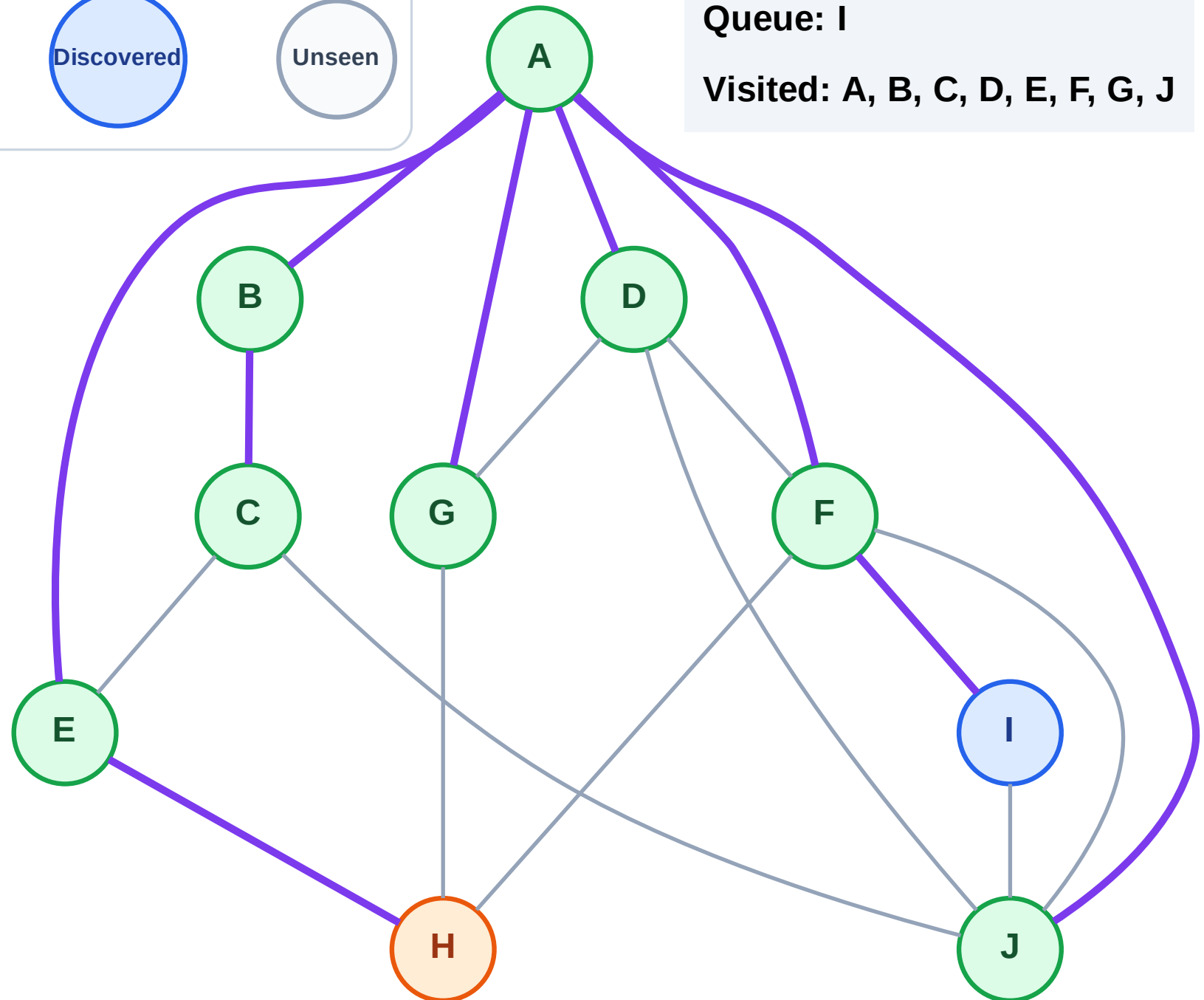
Step 18: Process node H

Legend



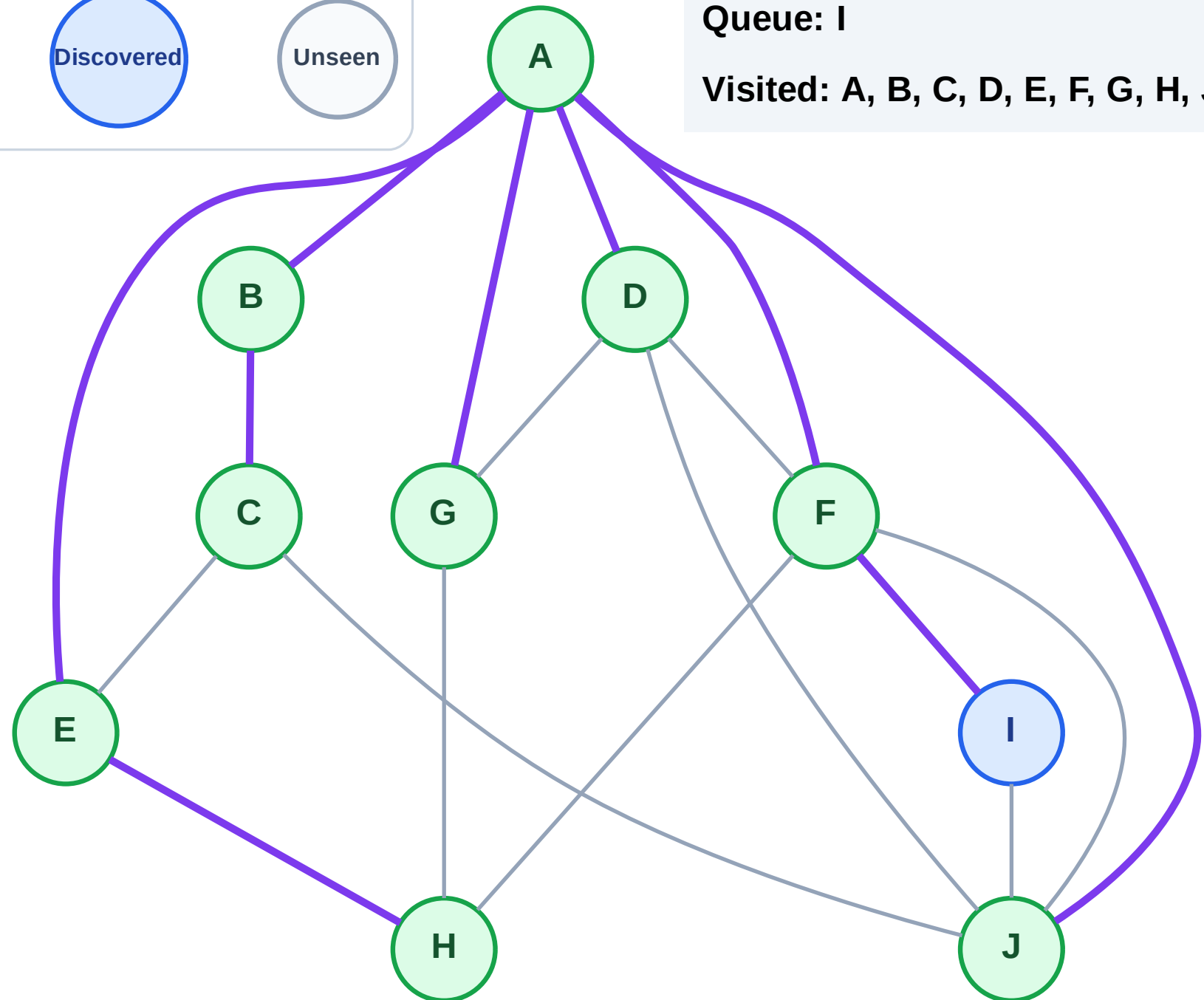
Queue: I

Visited: A, B, C, D, E, F, G, J



Step 19: No new neighbors from H

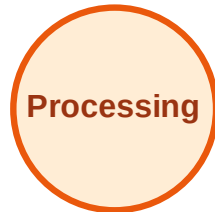
Step 19: No new neighbors from H



BFS Traversal

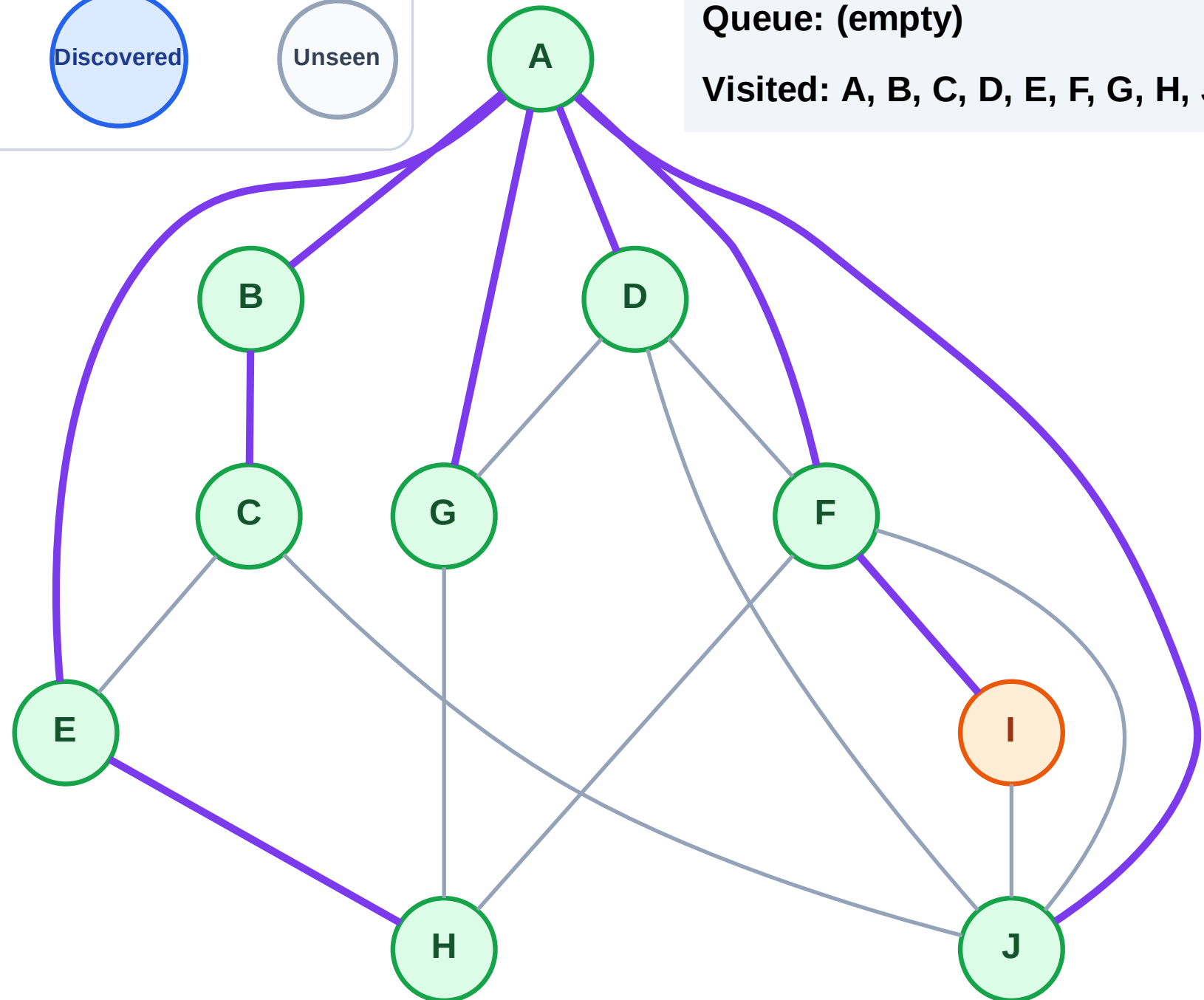
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

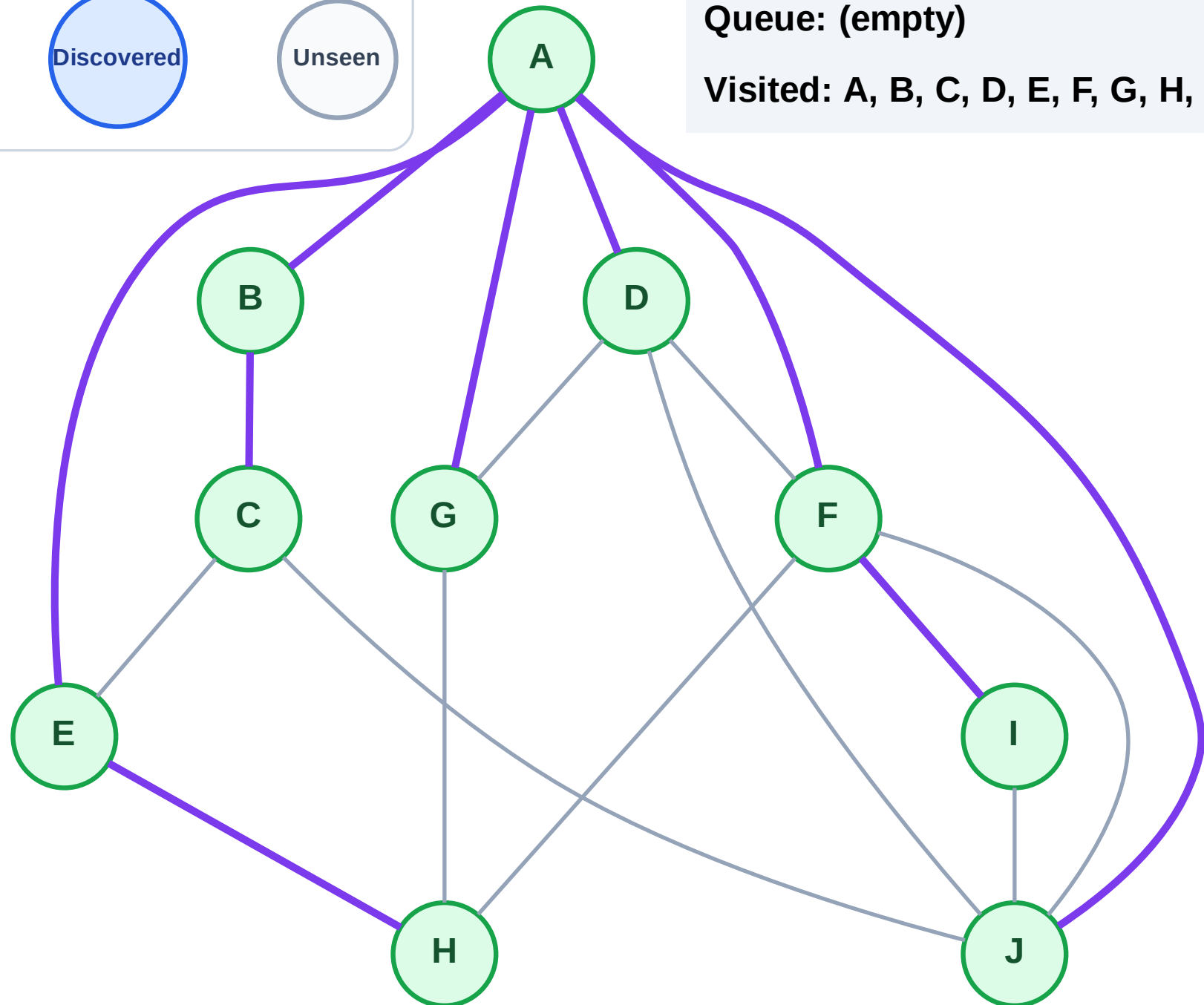
Step 21: No new neighbors from I

Legend



Queue: (empty)

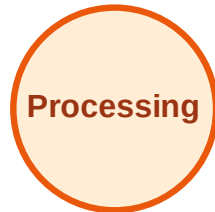
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

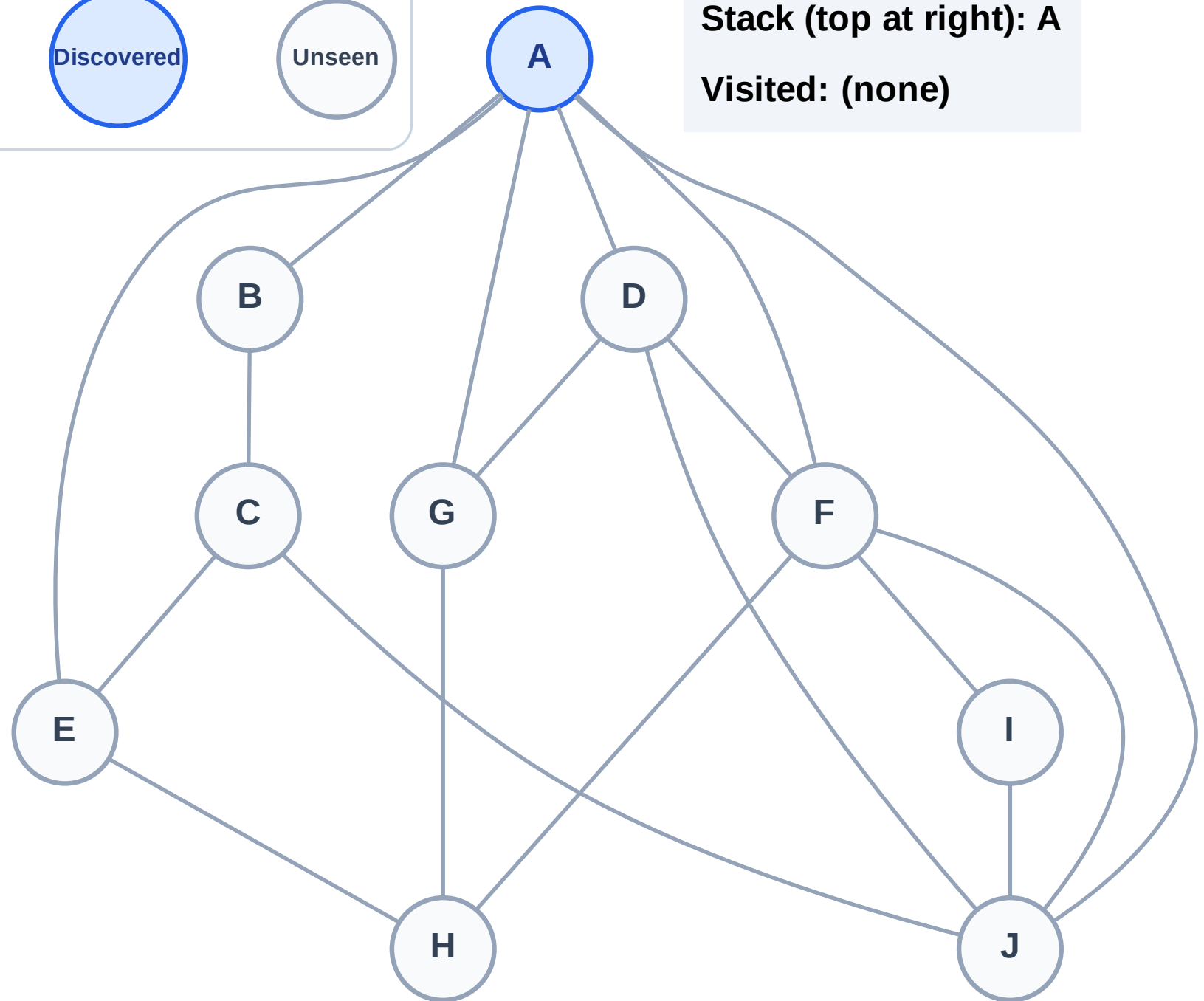
Step 1: Start at A

Legend



Stack (top at right): A

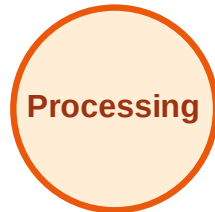
Visited: (none)



DFS Traversal

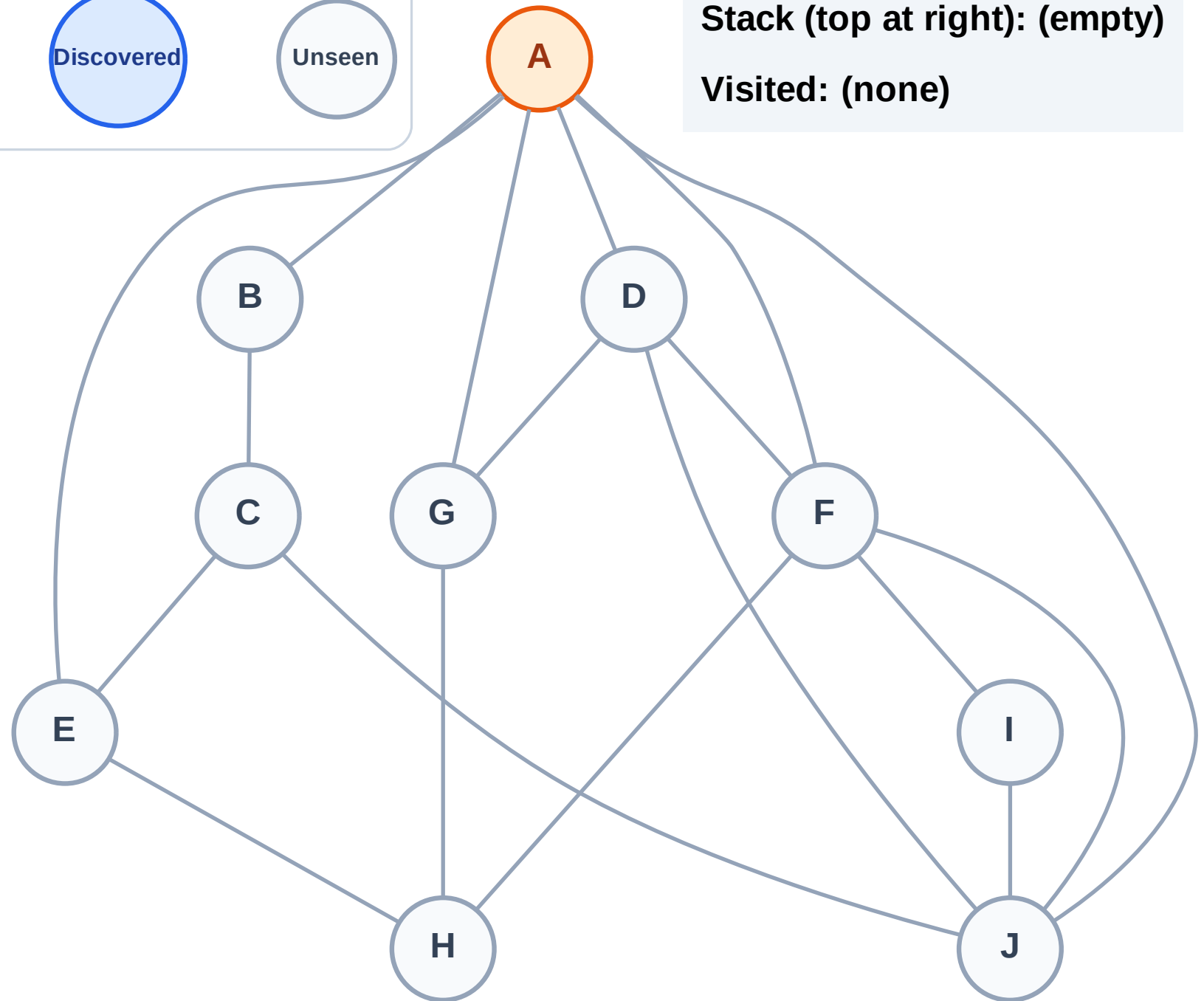
Step 2: Process node A

Legend



Stack (top at right): (empty)

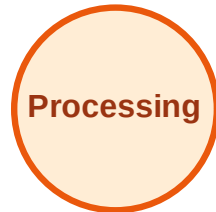
Visited: (none)



DFS Traversal

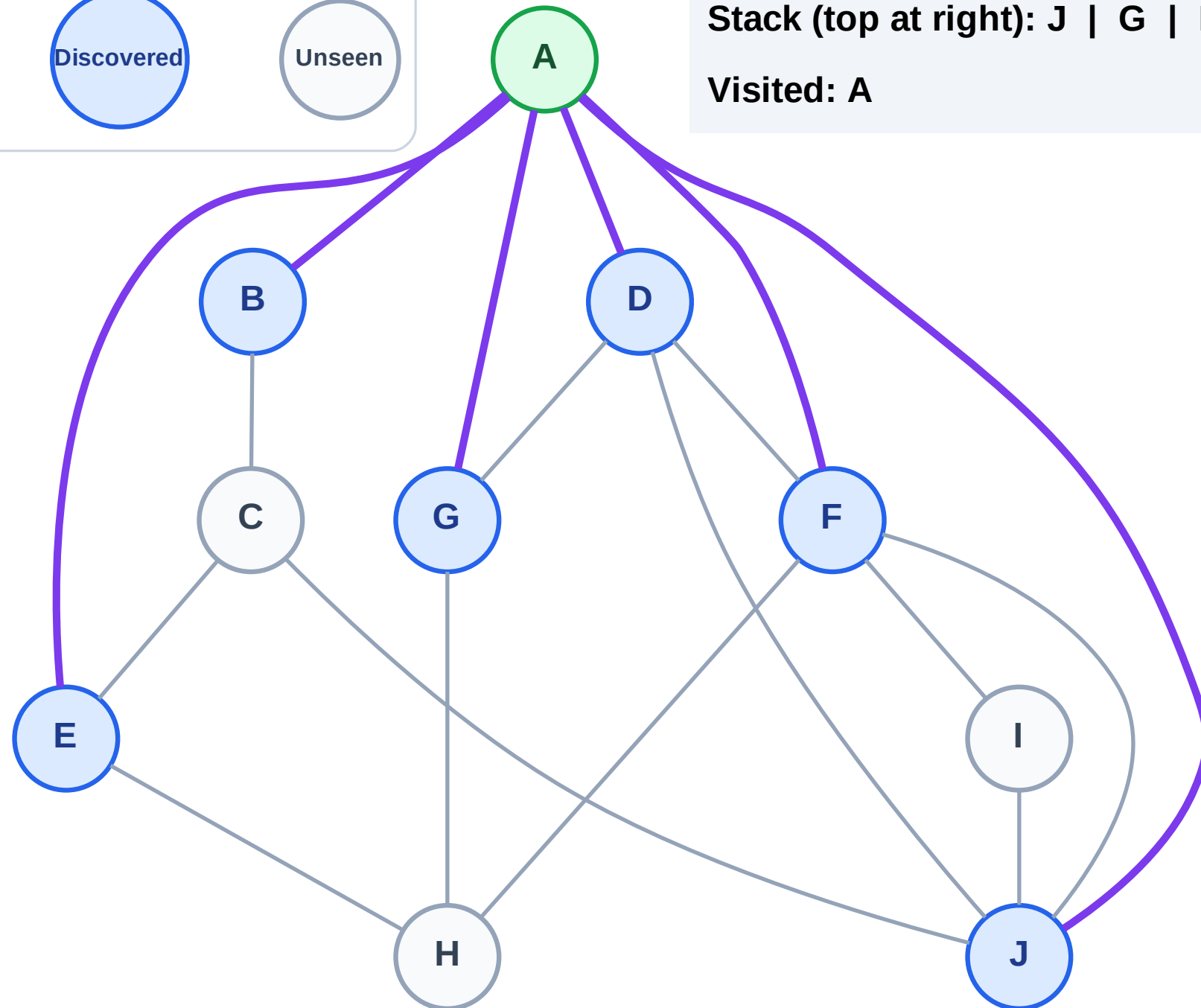
Step 3: Discover J, G, F, E, D, B from A

Legend



Stack (top at right): J | G | F | E | D | B

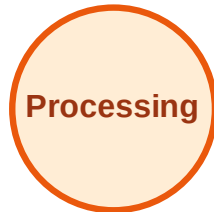
Visited: A



DFS Traversal

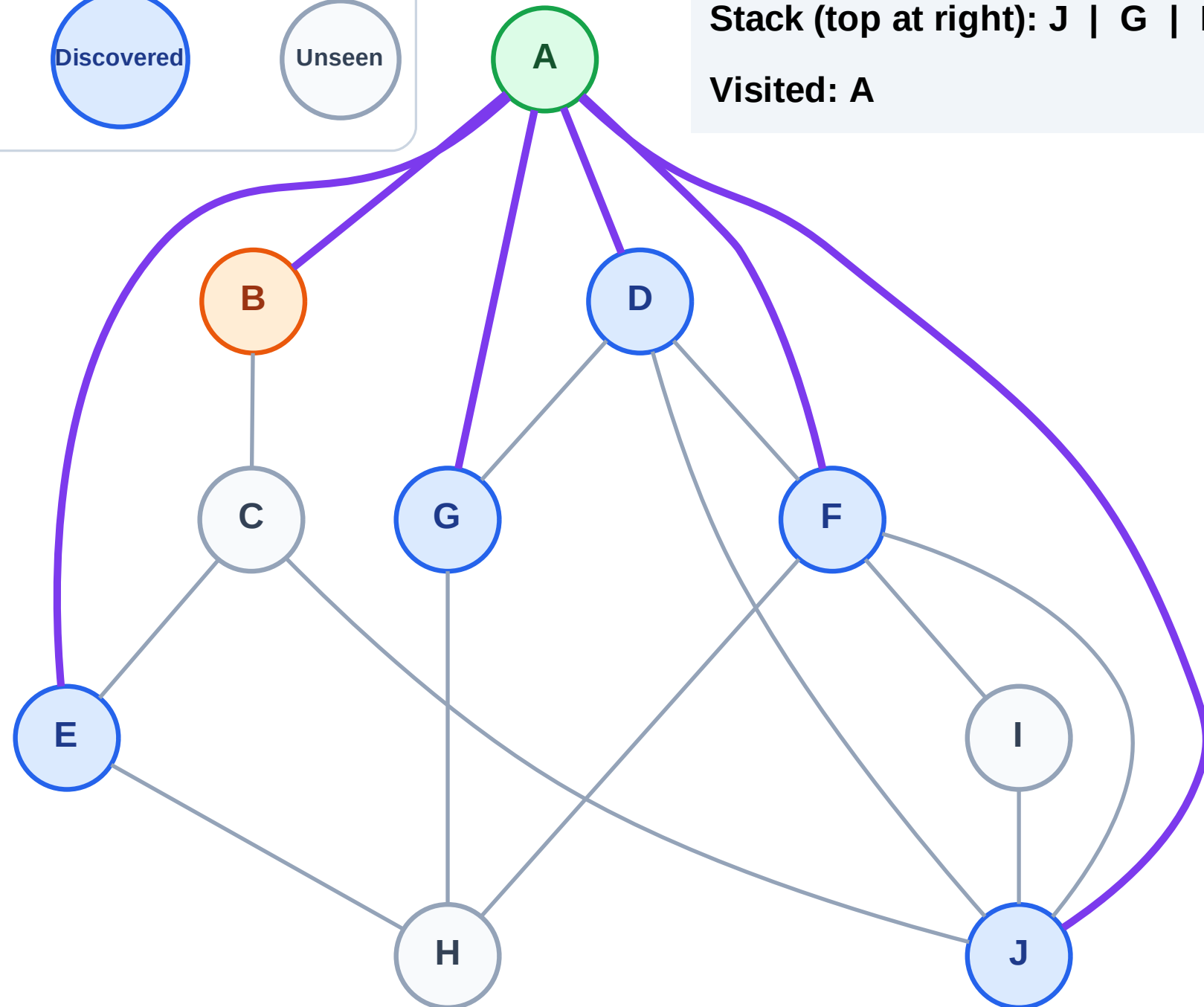
Step 4: Process node B

Legend



Stack (top at right): J | G | F | E | D

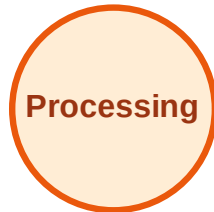
Visited: A



DFS Traversal

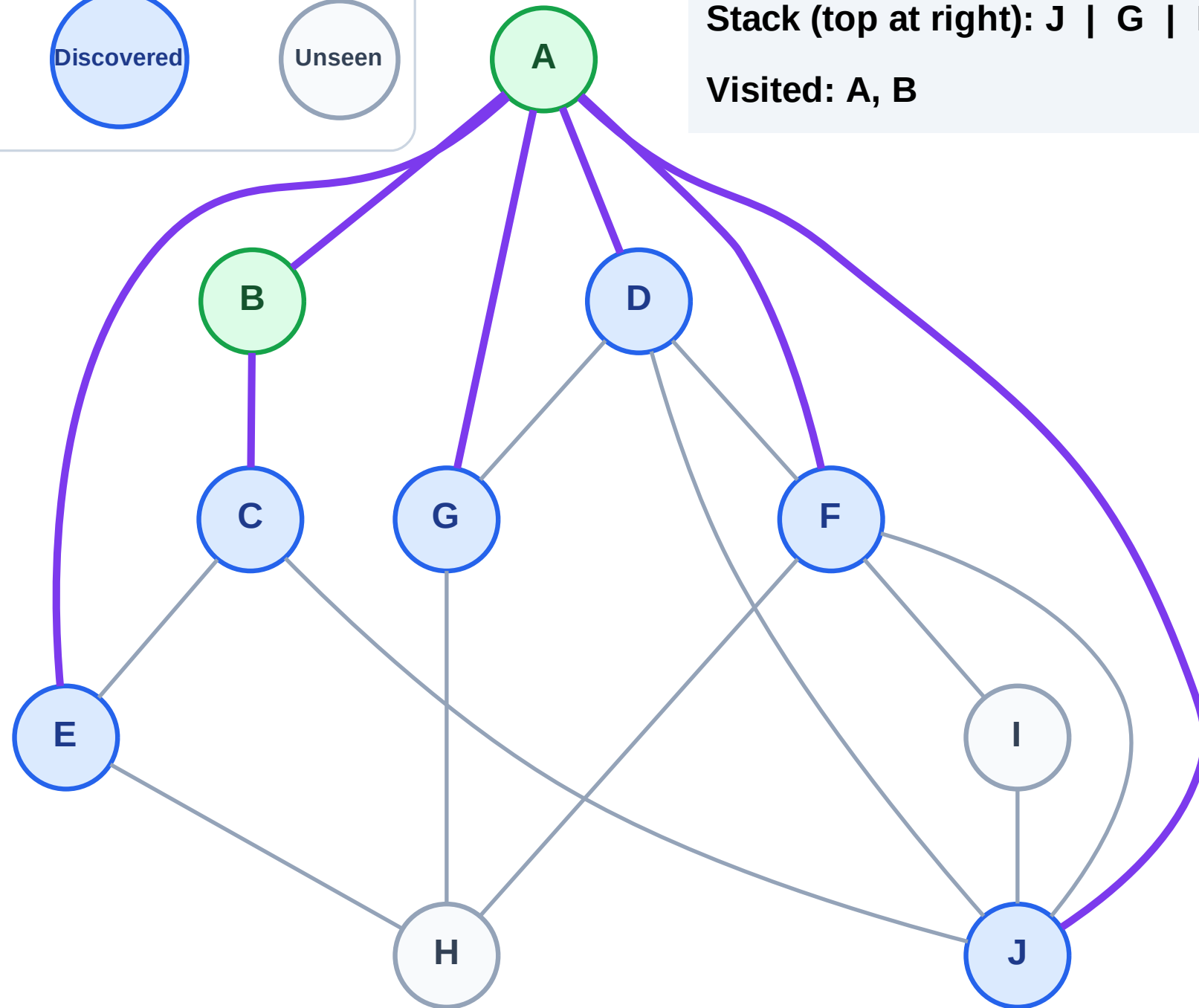
Step 5: Discover C from B

Legend



Stack (top at right): J | G | F | E | D | C

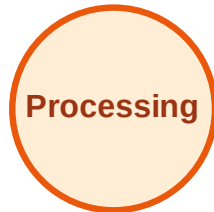
Visited: A, B



DFS Traversal

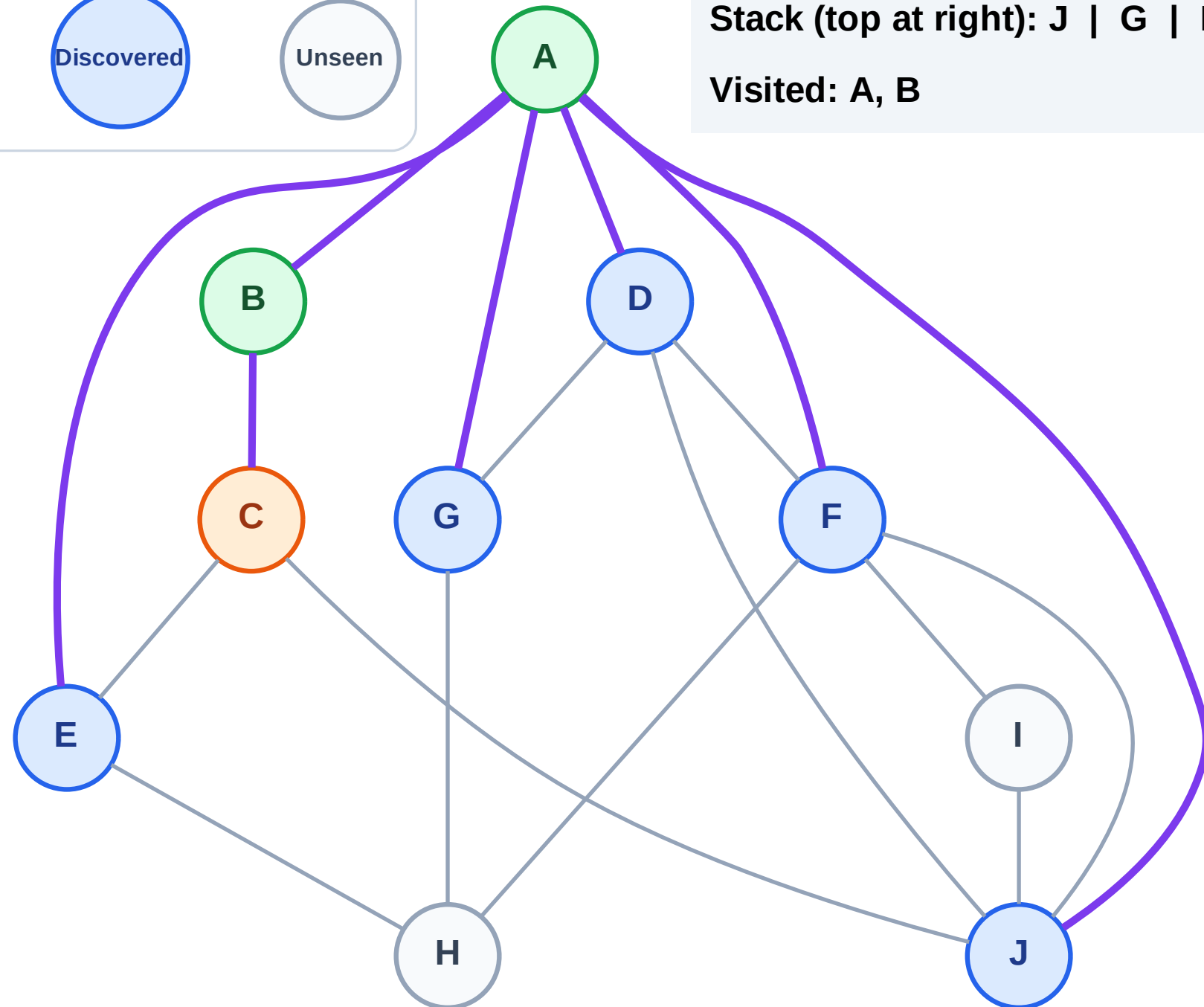
Step 6: Process node C

Legend



Stack (top at right): J | G | F | E | D

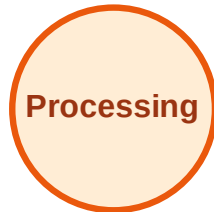
Visited: A, B



DFS Traversal

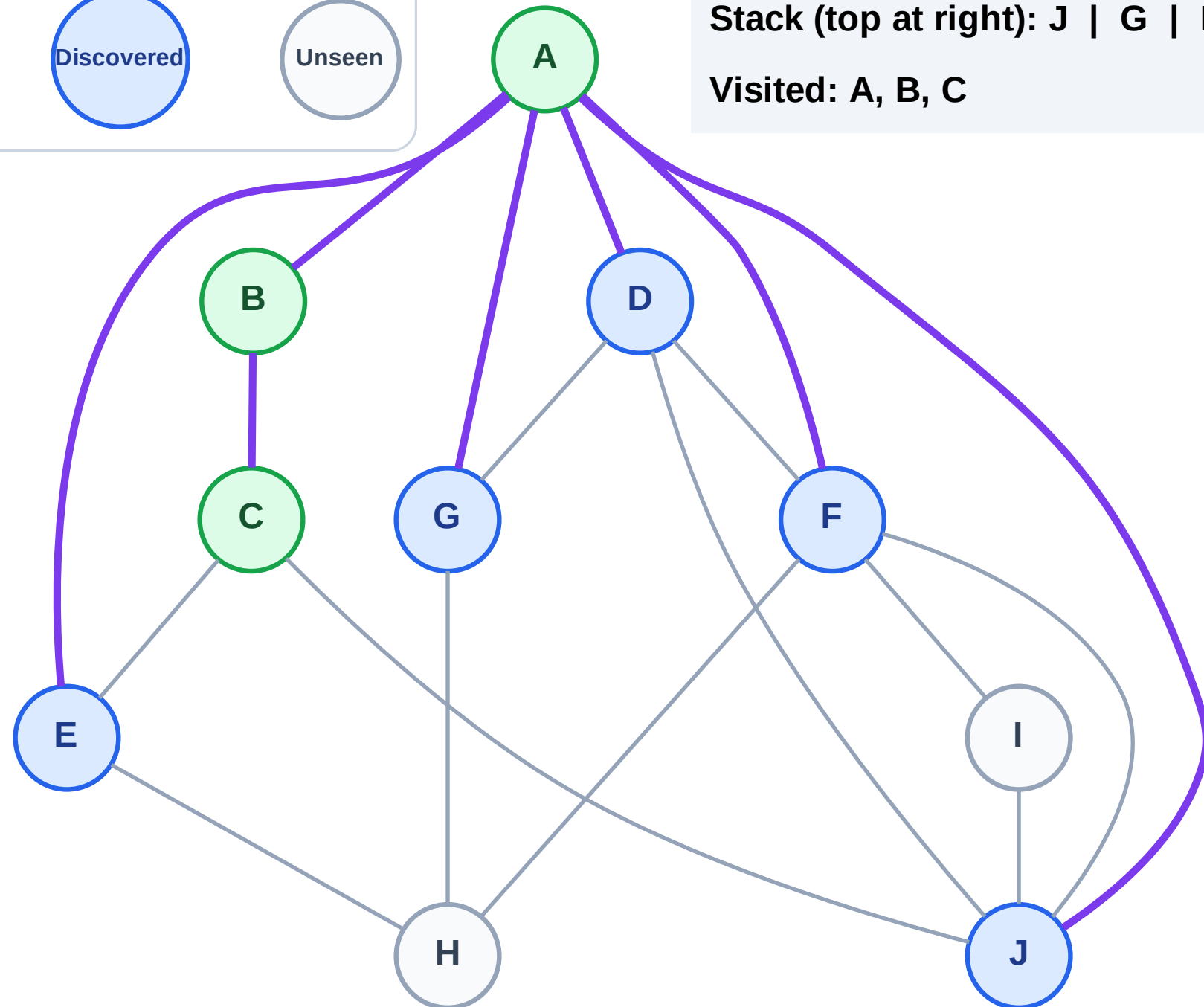
Step 7: No new neighbors from C

Legend



Stack (top at right): J | G | F | E | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

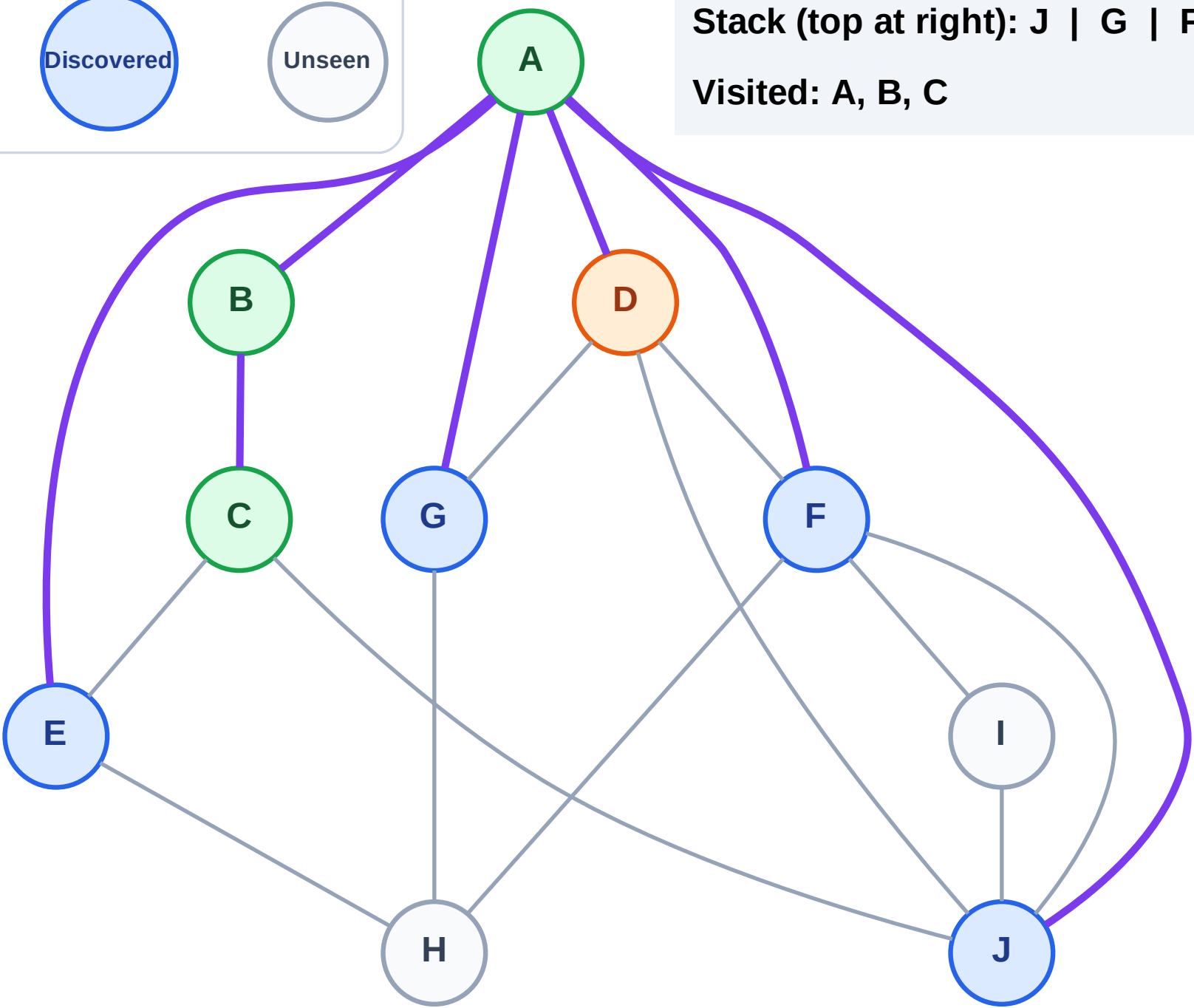
Processing

Discovered

Unseen

Stack (top at right): J | G | F | E

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from D

Legend

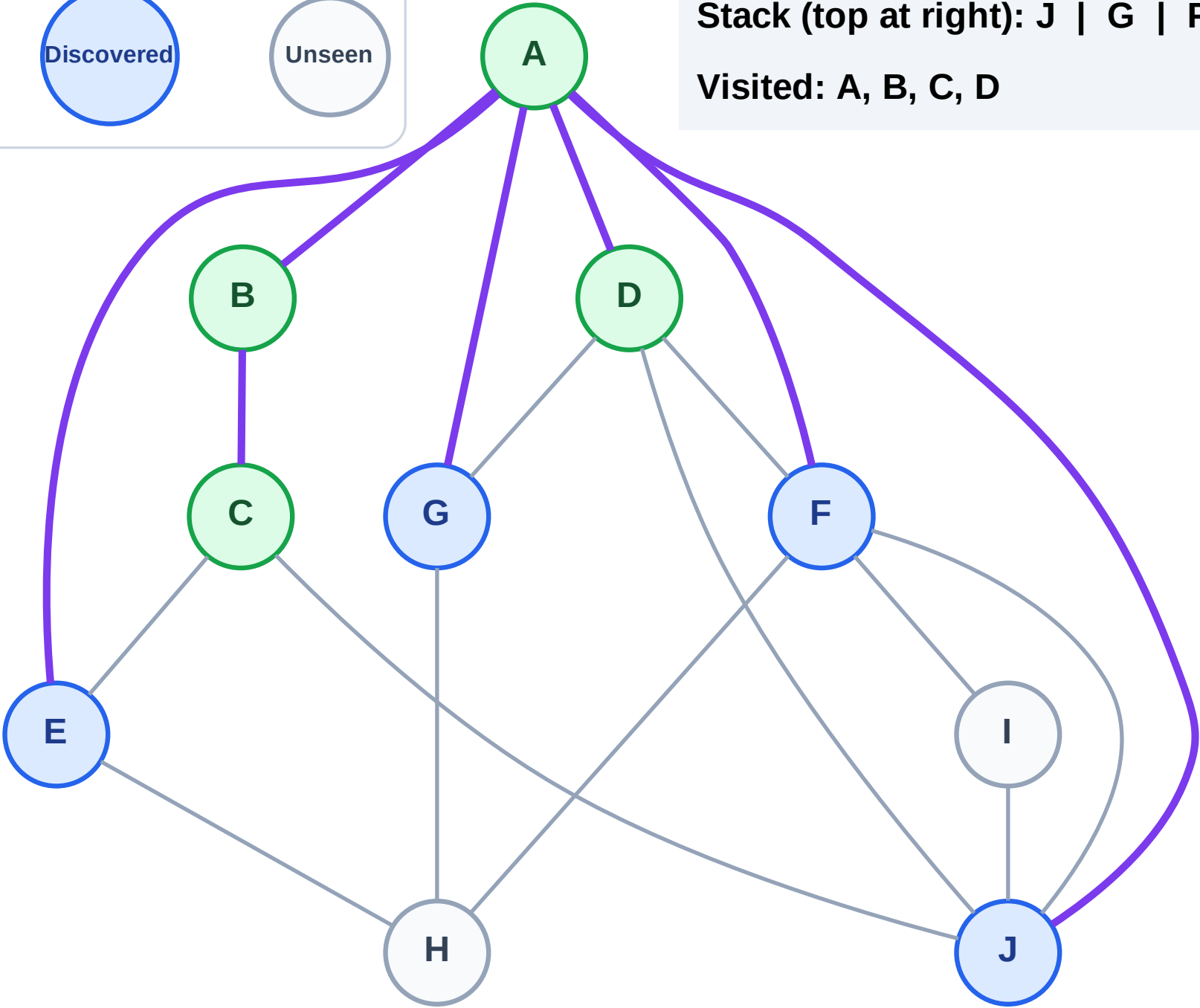
Complete

Processing

Discovered

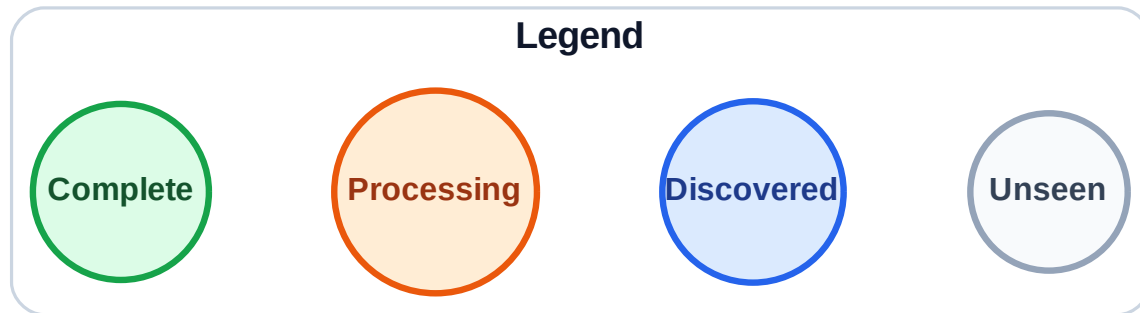
Unseen

Stack (top at right): J | G | F | E
Visited: A, B, C, D



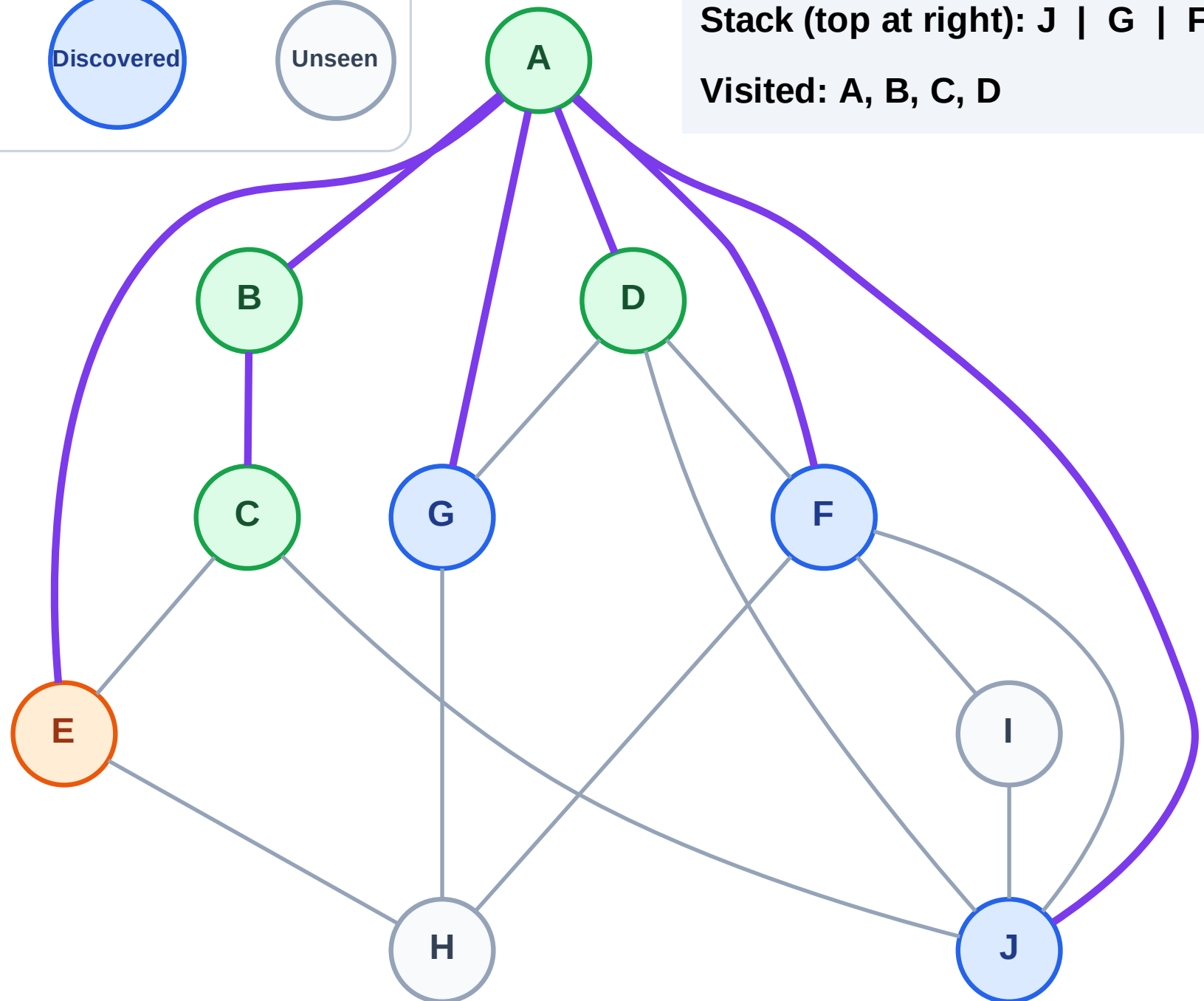
Step 10: Process node E

Step 10: Process node E



Stack (top at right): J | G | F

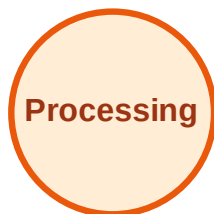
Visited: A, B, C, D



DFS Traversal

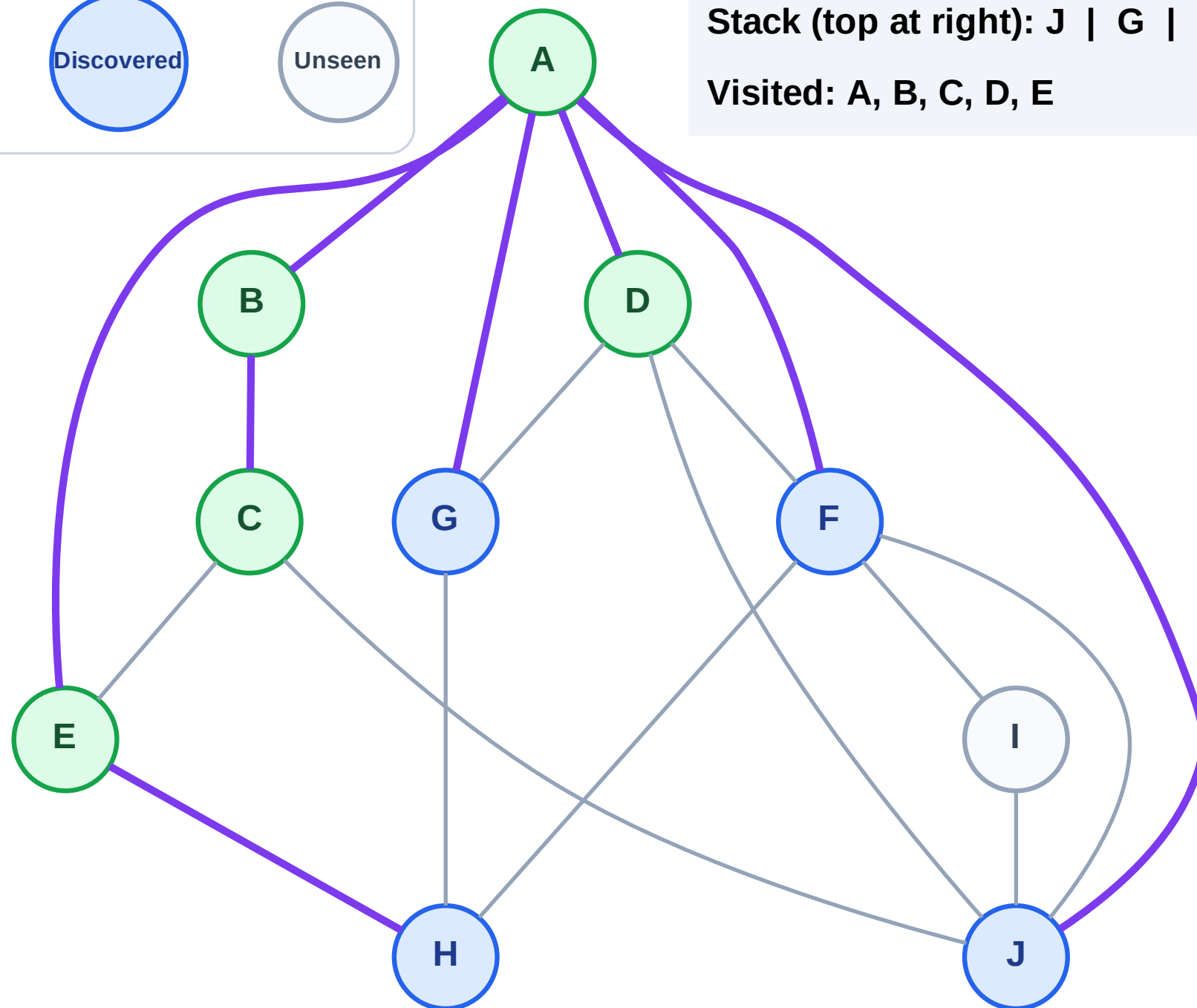
Step 11: Discover H from E

Legend



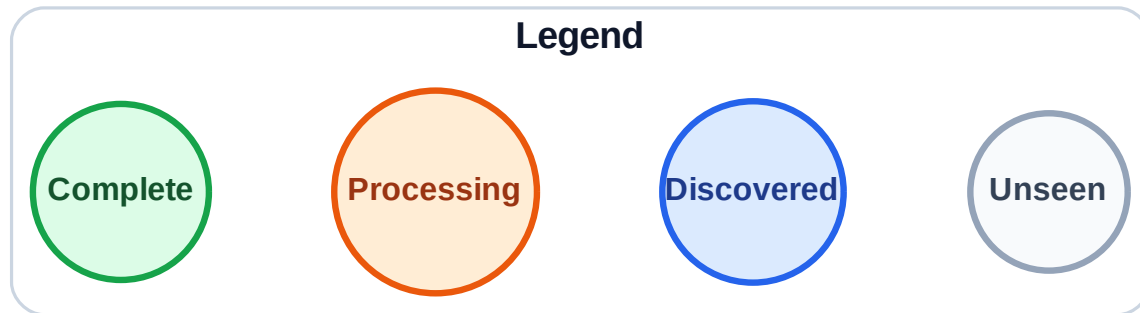
Stack (top at right): J | G | F | H

Visited: A, B, C, D, E



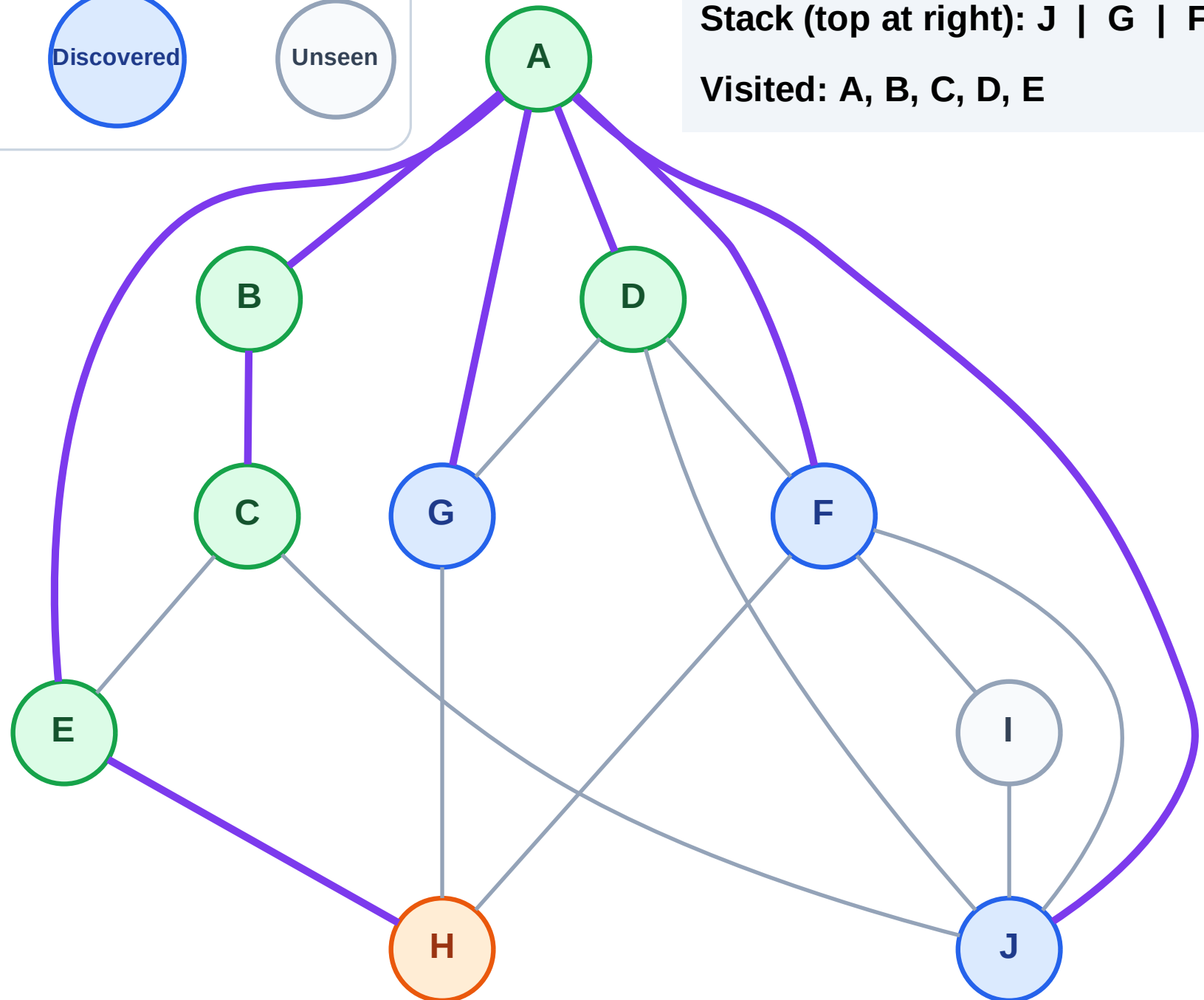
Step 12: Process node H

Step 12: Process node H



Stack (top at right): J | G | F

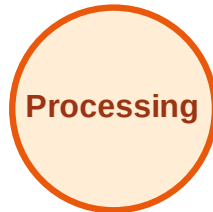
Visited: A, B, C, D, E



DFS Traversal

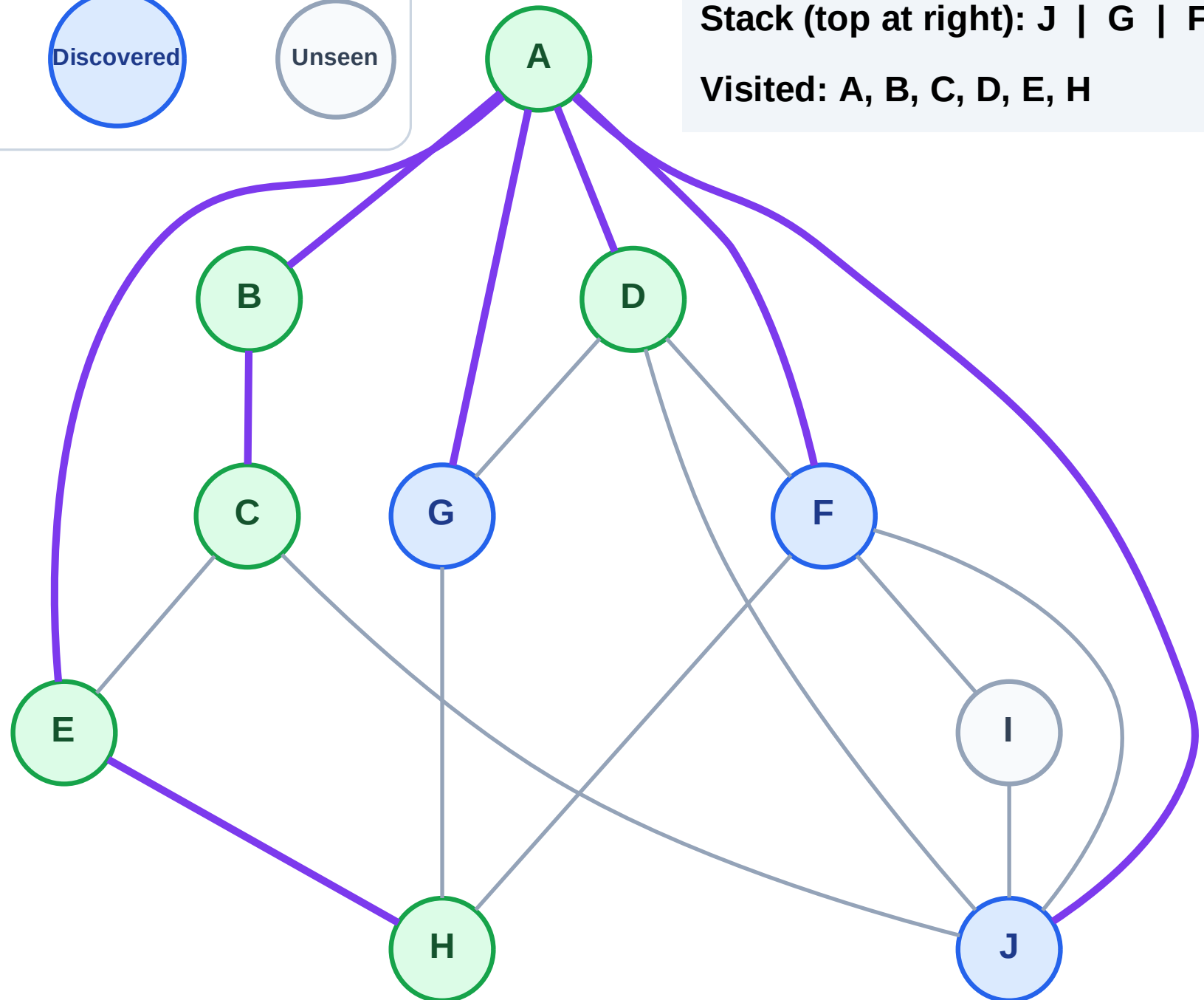
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | F

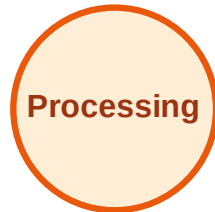
Visited: A, B, C, D, E, H



DFS Traversal

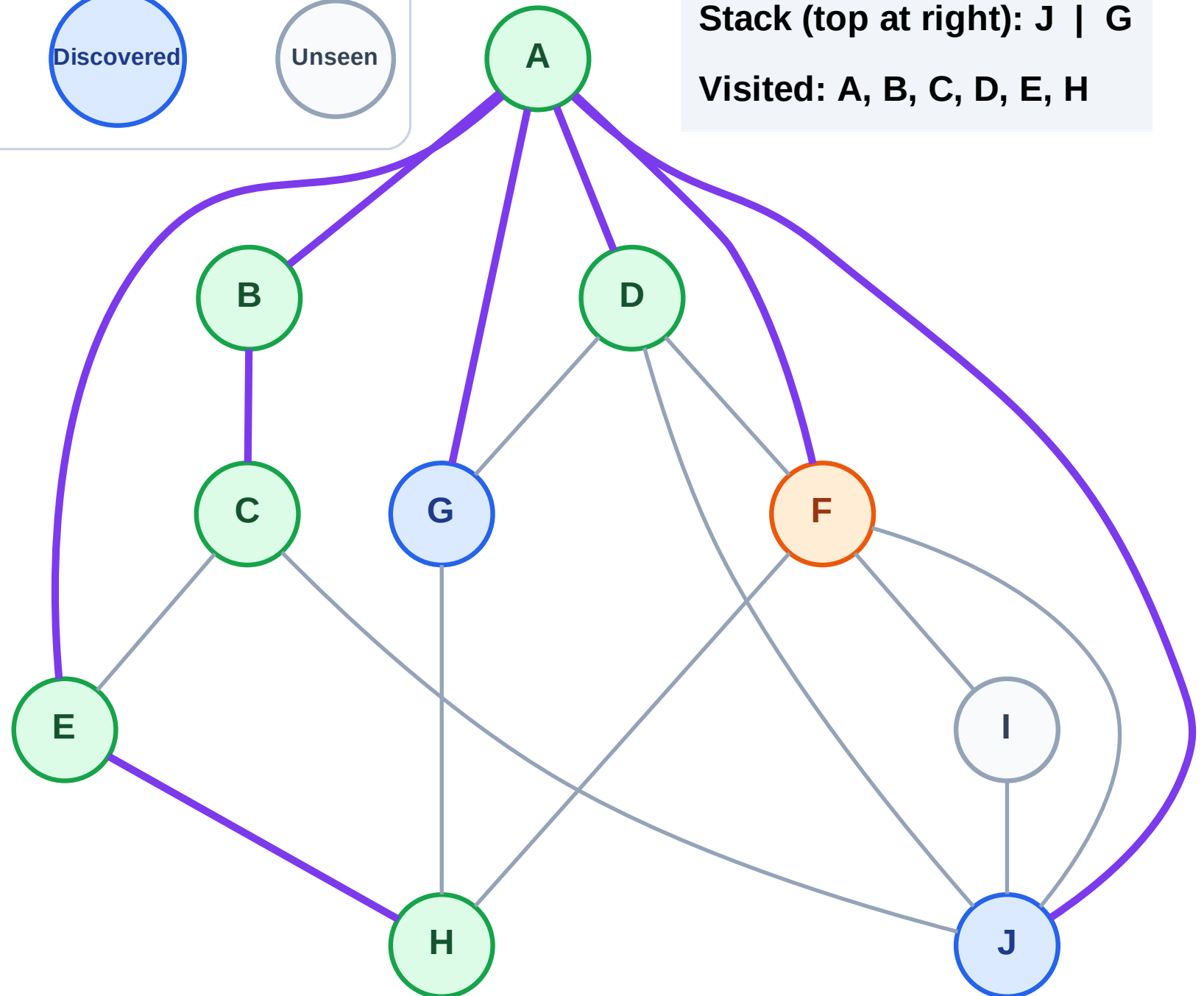
Step 14: Process node F

Legend



Stack (top at right): J | G

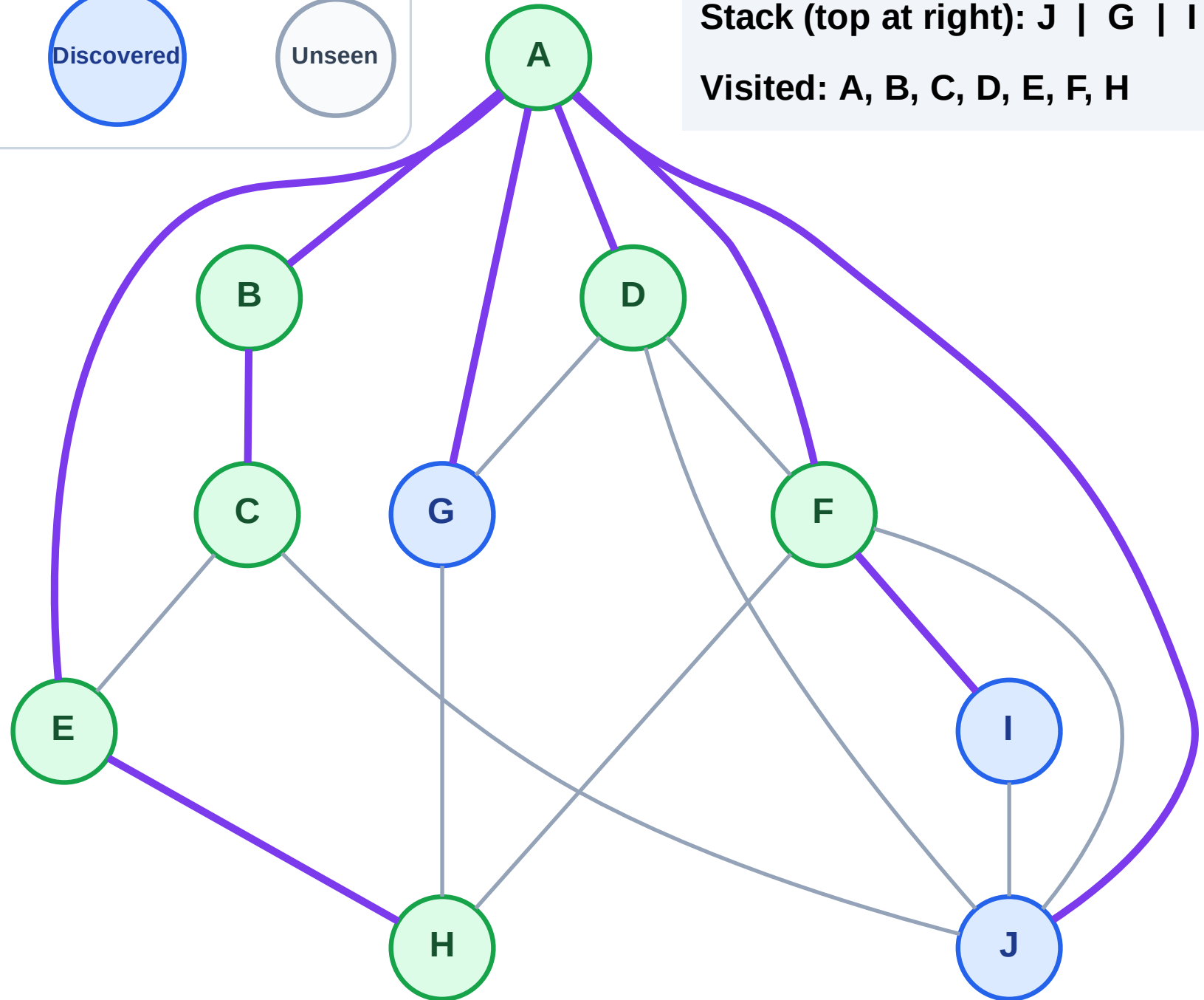
Visited: A, B, C, D, E, H



Step 15: Discover I from F



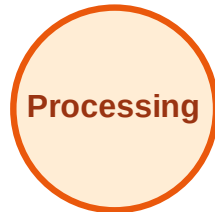
Visited: A, B, C, D, E, F, H



DFS Traversal

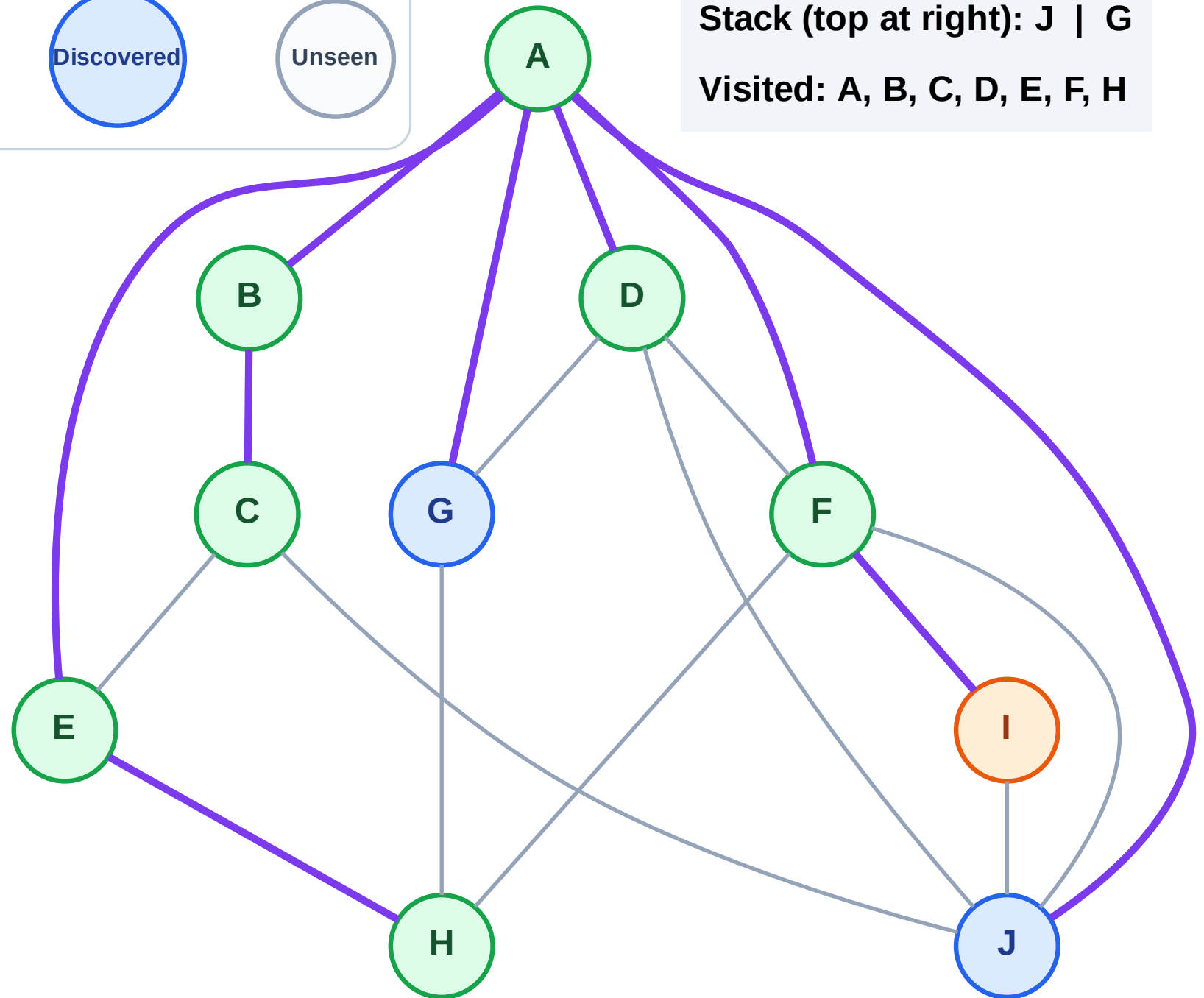
Step 16: Process node I

Legend



Stack (top at right): J | G

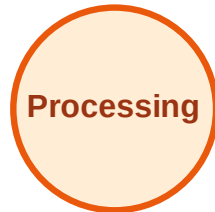
Visited: A, B, C, D, E, F, H



DFS Traversal

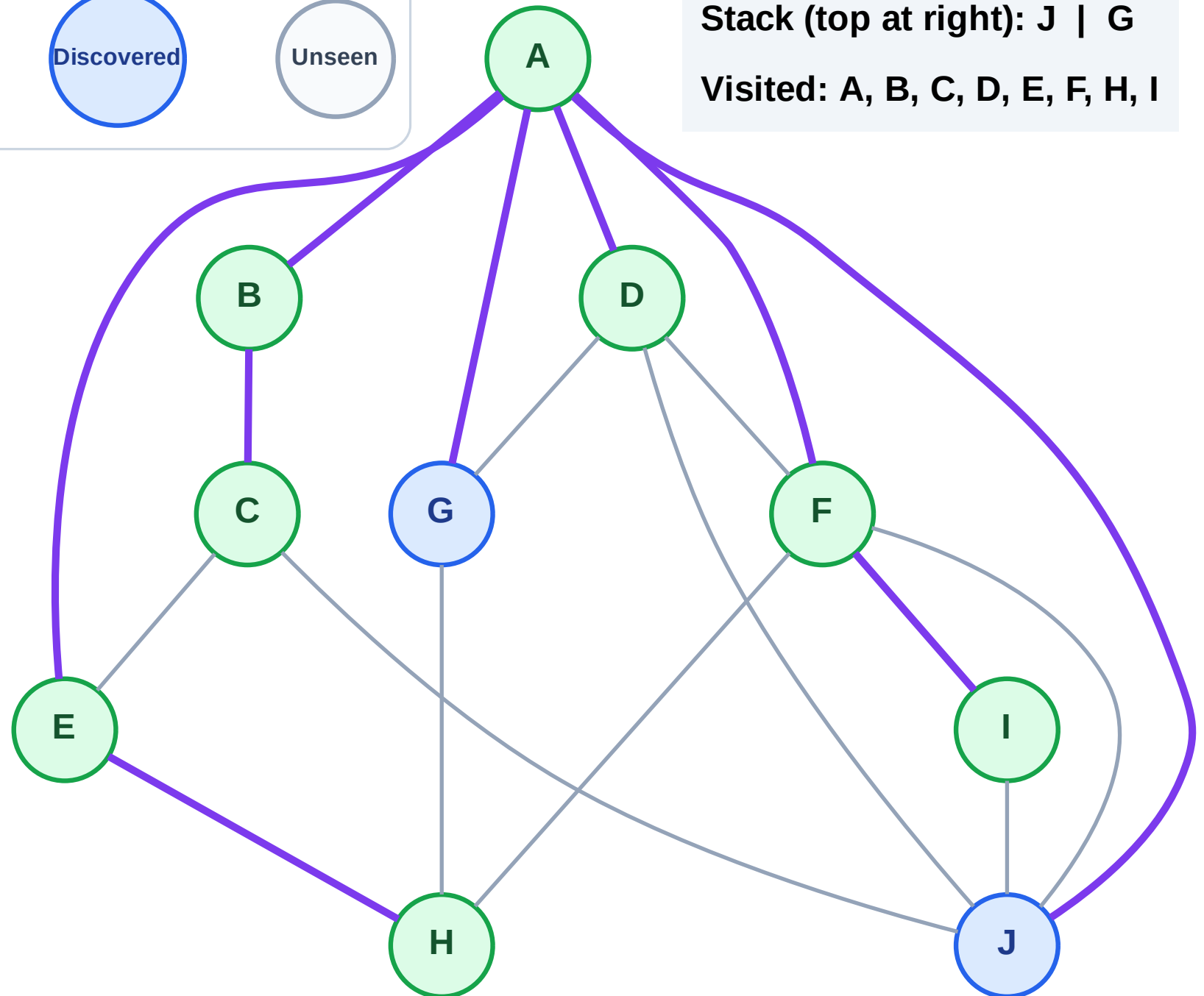
Step 17: No new neighbors from I

Legend



Stack (top at right): J | G

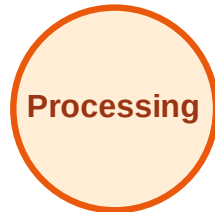
Visited: A, B, C, D, E, F, H, I



DFS Traversal

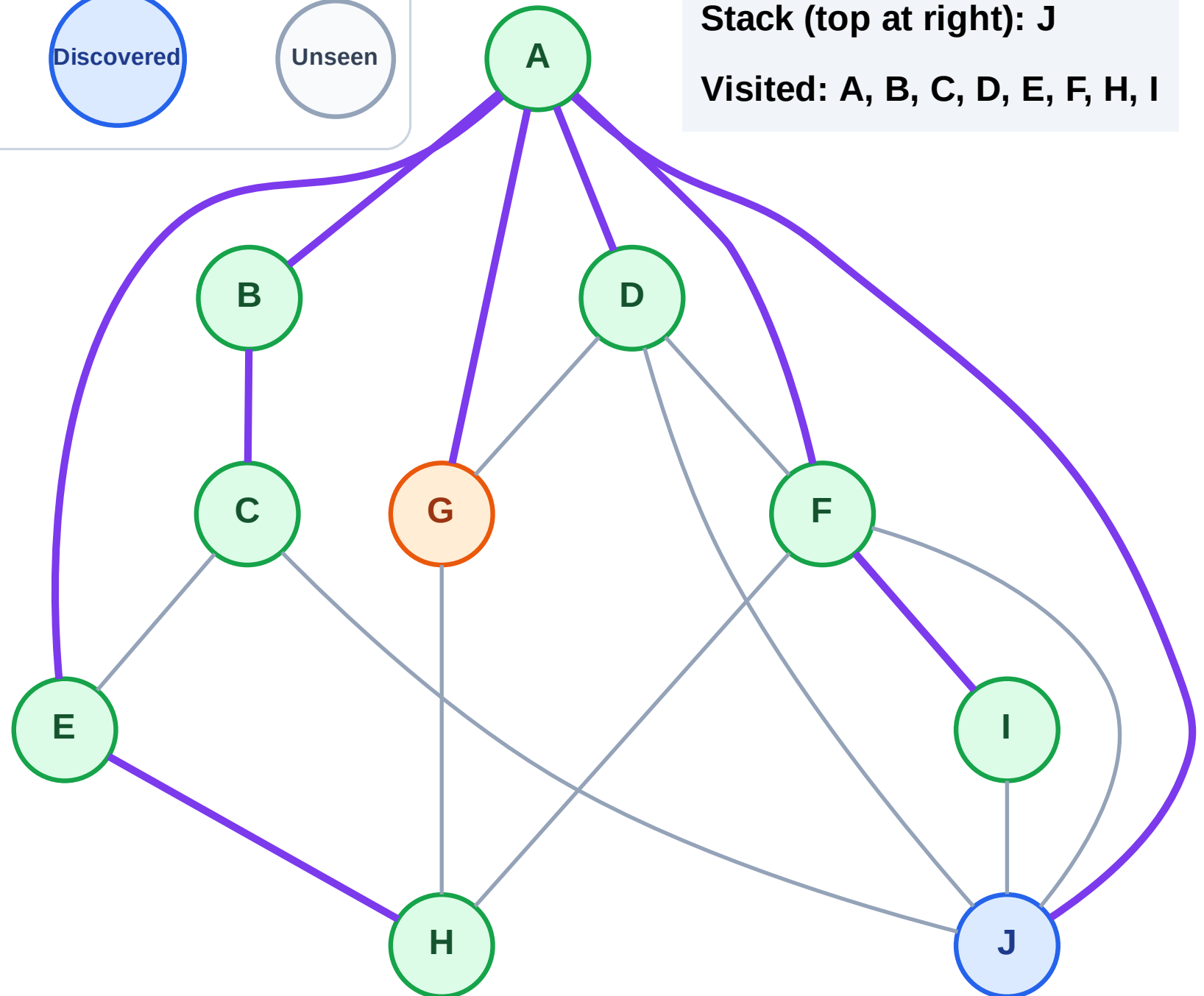
Step 18: Process node G

Legend



Stack (top at right): J

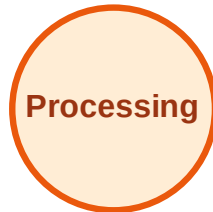
Visited: A, B, C, D, E, F, H, I



DFS Traversal

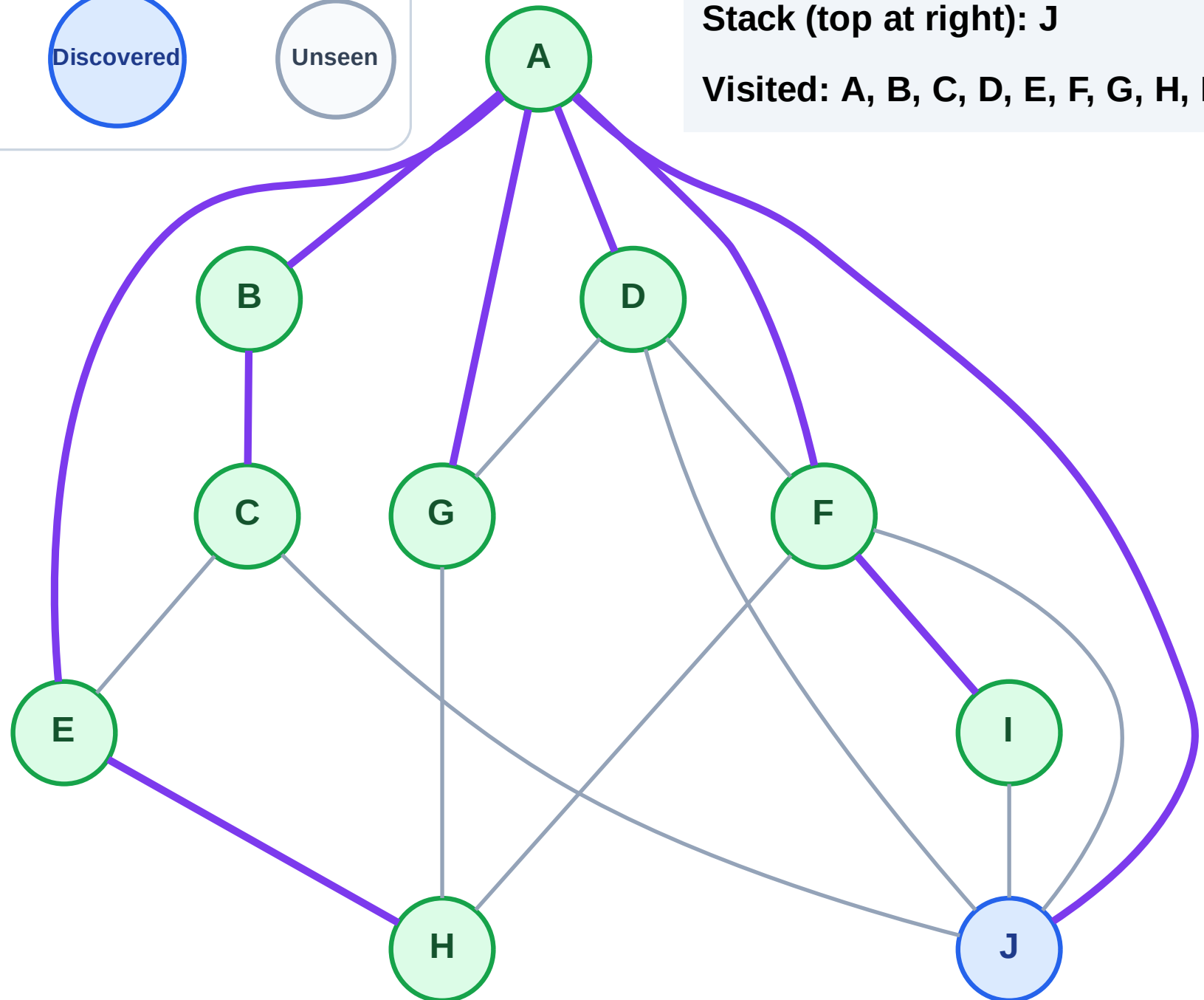
Step 19: No new neighbors from G

Legend



Stack (top at right): J

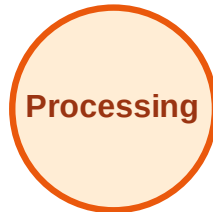
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

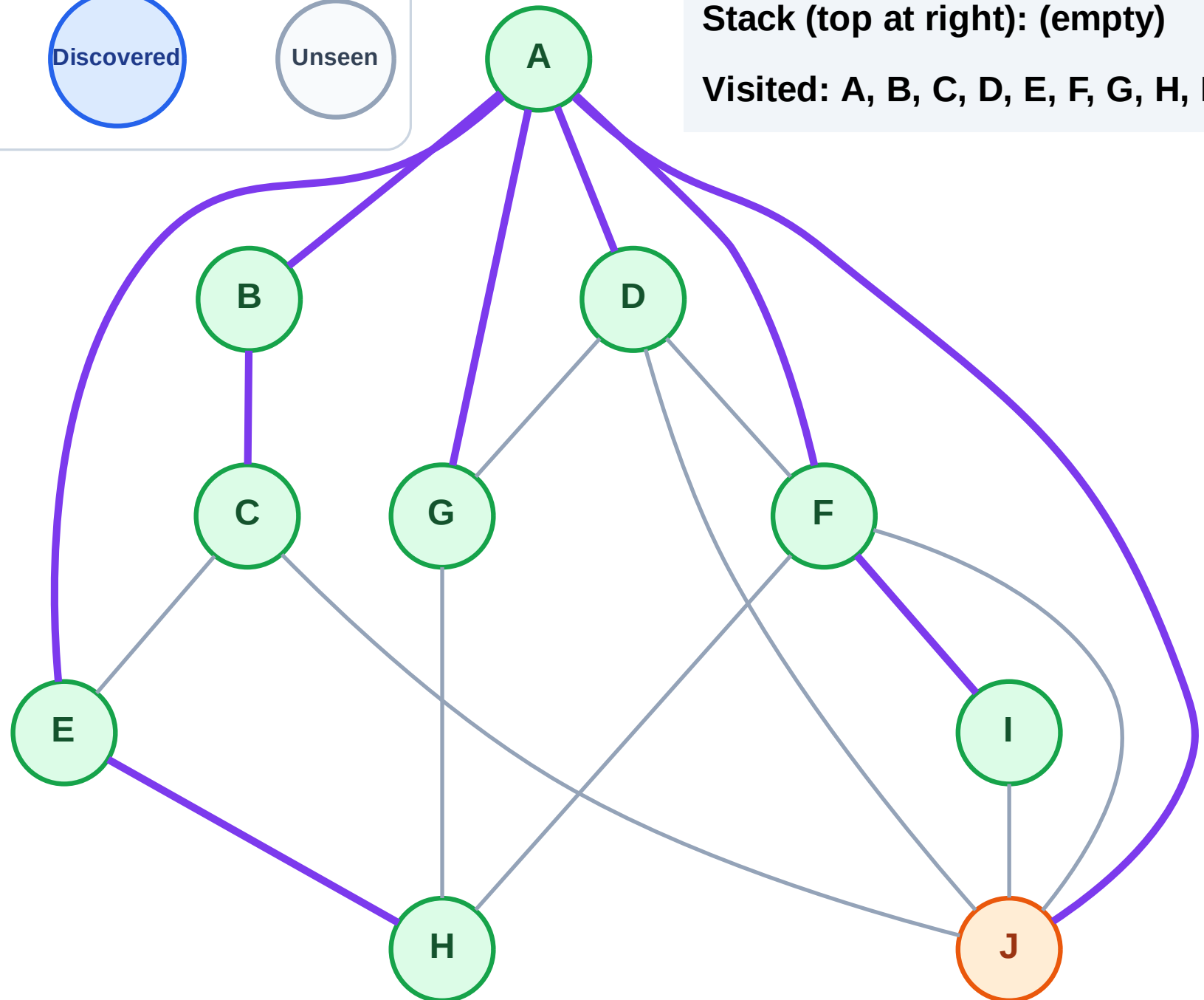
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

